

Australian National University

FIRM Model

Documentation

RE100 Group

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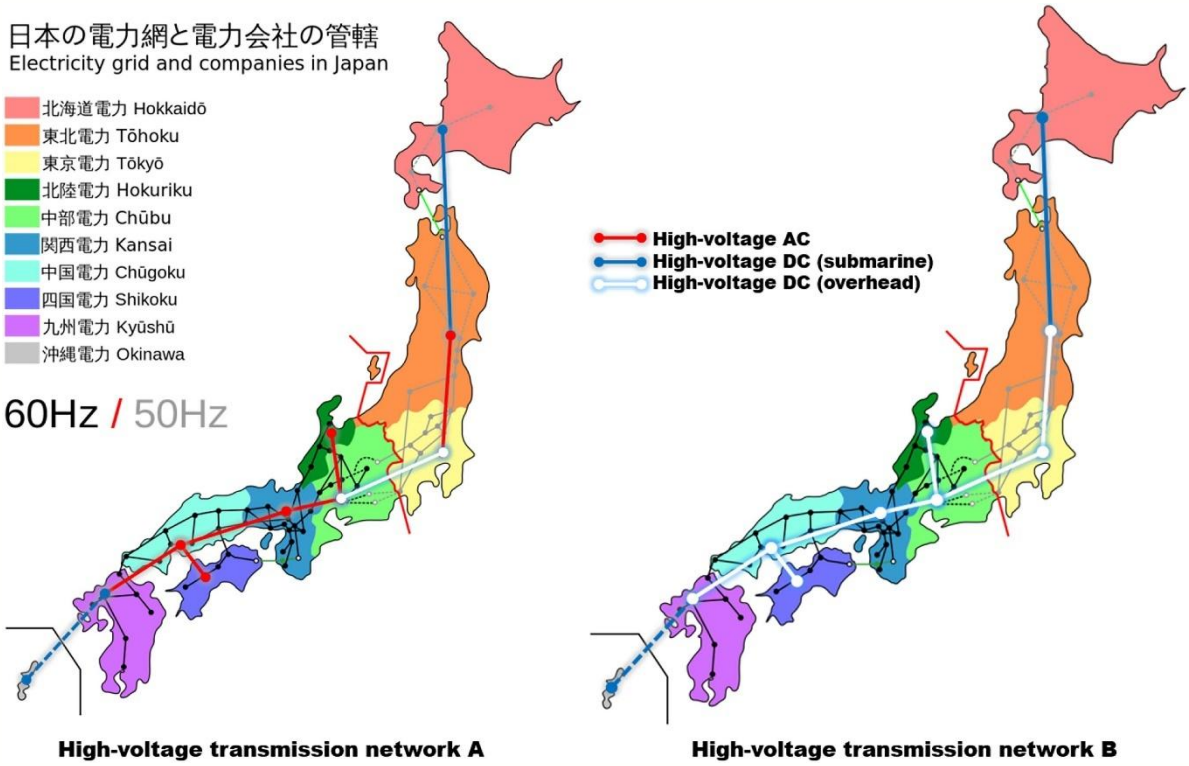
1 Introduction

The FIRM model determines the least-cost electricity system configuration for a pre-defined electricity network that meets certain reliability, resource, energy and transmission constraints. It uses a net-load approach to balance electricity demand with generation on an interval basis (typically 1-hour or 30-minute intervals) over a period of many years (typically 10 to 40 years). The model is used to determine upper bounds on the capacity of renewable generators and storage required for a 100% renewable electricity network (with the possibility of including legacy fossil fuel power stations). It also allows analysis of the levelised cost of electricity at a network-level to be performed.

FIRM simply refers to firmed electricity supply and is not an acronym.

The electricity network within the model is defined to be a set of interconnected nodes. An example for two hypothetical Japanese networks with high-voltage transmission between the major load centres is provided in Figure 1.[1] Each node indicates a location for inputs such as solar, wind, and electricity demand data to be added. The interconnections allow for generation to flow between nodes to meet demand at a different location.

Figure 1. Hypothetical Japanese electricity networks



[1] C. Cheng, A. Blakers, M. Stocks and B. Lu, "100% Renewable Energy in Japan," Energy Conversion and Management, vol. 255, no. 115299, 2022.

From the Supplementary Information B of Cheng, et al., (2022) the model constraints are defined to be:

- **Reliability constraint**: electricity generation must meet demand in every timestep unless a specified amount of deficit is allowed, to represent load shedding in certain scenarios.
- **Resource constraint**: installed capacity of a technology in a site must not exceed the identified technical resource potential of this technology in this site.
- **Energy constraint**: total generation from a certain technology must not exceed the specified maximum generation from this technology.
- **Transmission constraint**: power flow in a transmission line must not exceed the specified maximum capacity of this transmission line.

The decision variables for the least-cost optimisation are defined to be the following by default:

- Solar photovoltaic capacity at solar sites.
- Wind capacity at wind sites.
- Storage capacity at nodes.
- Power imported from external nodes; and
- Flexible hydrogen capacity at nodes.

Scenario analysis with the FIRM model enables the impact of changes to the network design or input assumptions to be performed. Some standard scenarios that can be analysed include:

- Changes in electricity demand per capita per year;
- Changes in the shape of the daily demand curve;
- Adding or removing interconnections;
- Modifying the cost assumptions; and
- Adding or removing types of firming capacity (e.g., hydrogen).

1.1 Assumptions

Simplifying assumptions are made to improve the optimisation speed of the model or make it easier to develop the system.

- The **copper plate** assumption means that electricity can flow unconstrained from generators to loads. Some constraints could still be applied within the model to limit capacity of specific interconnections, though it is important to understand that this would result in a higher cost solution.
- Levelised cost of electricity is calculated at a network level, not a generator level.
- The least-cost dispatch is based on minimising costs from the perspective of the grid system operator and does not account for generators looking to maximise profits in environments with low competition (e.g., high-offering price makers in an energy pool market when baseload generation is low).
- Solar PV, wind, and flexible hydrogen are, in general, the sole source of new generation capacity. In 2021, solar and wind constituted 75% of new global capacity installations. At current growth rates, it is reasonable to expect solar and wind will dominate new generation installations globally going forwards. There are currently no significant nuclear, carbon capture and storage, geothermal, or solar thermal developments at global scale, and it is highly unlikely that these industries will compete with solar and wind by 2050. There is also unlikely to be significant biomass generation compared to solar and wind since it requires 20X the land as solar photovoltaics to generate the same amount of electricity. Hydroelectric generation requires damming of rivers, so sites for power stations are rapidly becoming more limited. So, this assumption is reasonable when considering upper bounds on solar, wind, and hydrogen capacity in a 100% renewable electricity network.

"*All models are wrong, but some are useful*" - George Box

No models contain all of the complexity of reality. The FIRM model is no different. It is important to understand the underlying mechanics of the model and implications of the assumptions that have been made.

For example, the FIRM model will not contain all possible technologies that contribute to the energy balancing of the network (e.g., it may be missing demand management or future gas generators). This means that the capacity of solar, wind, and storage will not be a perfectly accurate representation of the future electricity network, but that does not mean that the results are useless. We can understand that the presence of demand management or additional firming capacity will reduce the need for variable renewable generation and storage, since these other technologies will fulfil the same generation or balancing role. Therefore, the outputs from the model can be considered **upper bounds**, since the model operates on a least-cost basis and the presence of these other technology solutions would be competitive with solar PV, wind, and storage because they are cost competitive.

Implications of the model results are also entirely dependent upon the assumptions that have been made. If the solar data we input had an anomalous cloudy period that lasted several weeks, the storage requirements would dramatically increase. If the cost of hydroelectricity was assumed to be significantly higher than the cost of hydrogen, then hydrogen would be dispatched for firming first which would not accurately describe the likely behaviour of the system. **Understanding the mechanisms** that lead to the results is essential for determining whether the model results are appropriate when making claims about the real world.

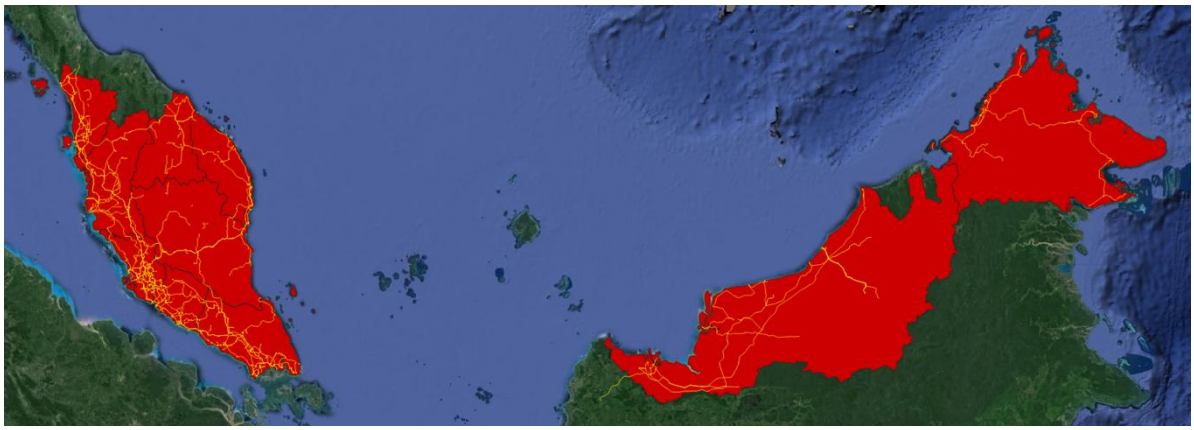
2 Inputs and Outputs

2.1 Electricity Network

The first step in developing a FIRM model for a particular scenario is to design the network of interconnected nodes that will be analysed. One very simple way of doing this is to look at the major load centres of the existing transmission network and define nodes at each of these locations. Interconnections can then follow the major existing transmission interconnections. This method is dependent on the availability of good quality transmission network data for the geography that is being analysed.

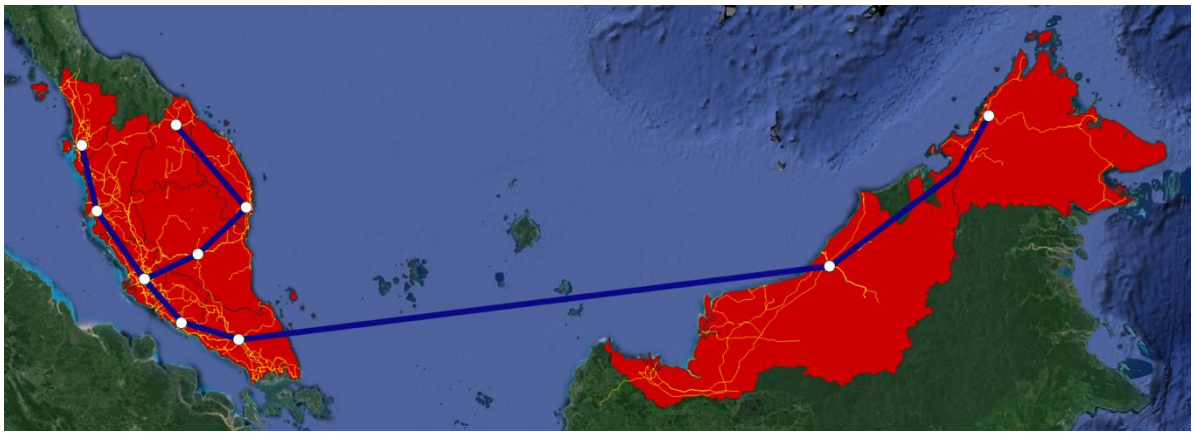
Open Street Maps includes transmission network data under the `power=line` tag. Distribution networks are also mapped using the `power=minor_line` tag, while underwater or underground cables are mapped with the `power=cable` tag. Figure 2 contains the transmission network data for Malaysia. The data has been queried using the QuickOSM tool for QGIS and is overlayed on a map of hypothetical service areas defined using the shapes of the states and territories. The image uses the Open Street Maps base map.

Figure 2. Open Street Maps transmission data (power=line) for Malaysia



Load centres have a high density of transmission interconnection and are typically located around major cities (e.g., Kuala Lumpur). A hypothetical high voltage backbone could be overlayed onto the map by drawing interconnected lines between the major sites forming these load centres. An example is provided in Figure 3.

Figure 3. Hypothetical high-voltage DC network based upon visual inspection of load centres

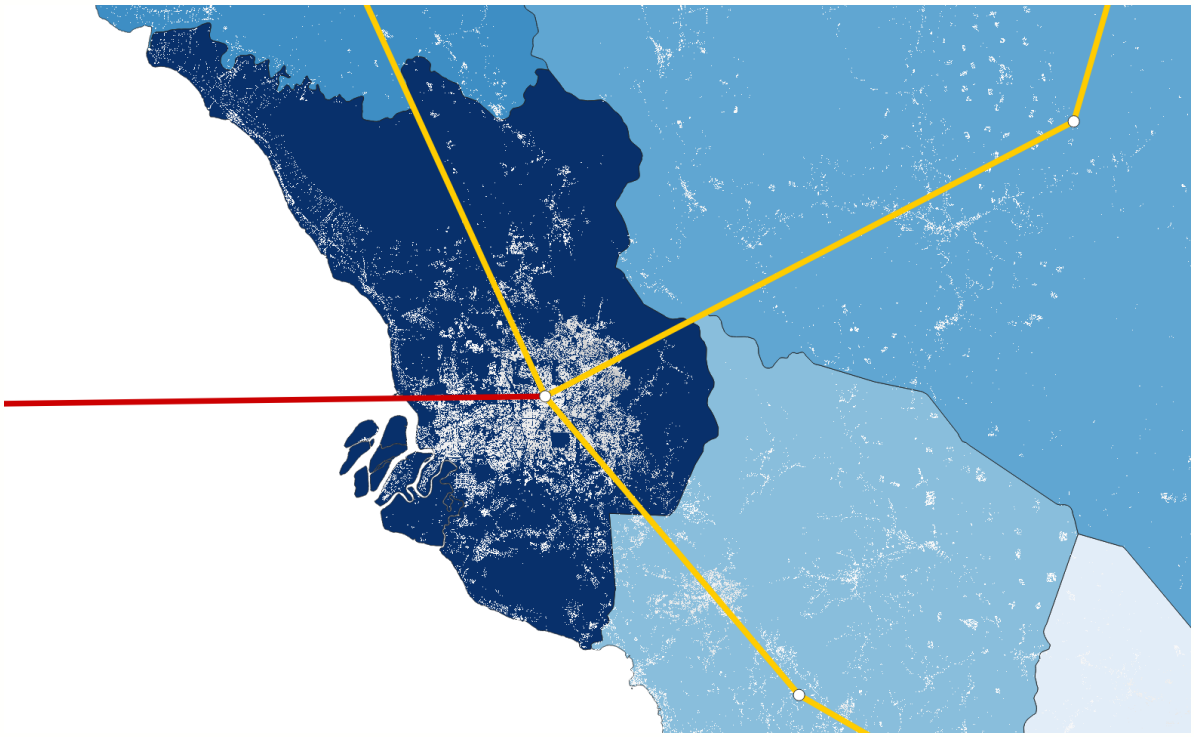


A more complex method of locating appropriate load centres could be based on population density across the geography. Loading a population density raster for the geography into GIS software, then finding the population-weighted centre of each service area, will give an indication of optimal node locations across the

geography.[2] This method assumes that the size of the load in a given location is proportional to the population in that area. The method may not be appropriate if there is large industry located far away from the population centres in each service area.

The location of a node based upon the population-weighted centre of the Selangor-Kuala Lumpur-Putrajaya service area is shown in Figure 4.

Figure 4. Selangor-Kuala Lumpur-Putrajaya node location based on population-weighted centre of service area



The FIRM model uses a copper plate assumption, so the least-cost network should simply define interconnection in a way that results in the shortest transmission network. This will minimise transmission losses across the network, while the copper plate assumption still allows a load to be balanced by generation at the furthest node (since transmission capacity is unconstrained, unless a constraint is explicitly defined in the model). Figure 5 describes the shortest electricity network for all of Malaysia based upon the population-weighted centres of each service area.

Each node and each interconnection should be assigned a unique identifier (e.g., a string that is several characters long). The node identifiers should be added as column headers to each of the following CSV files:

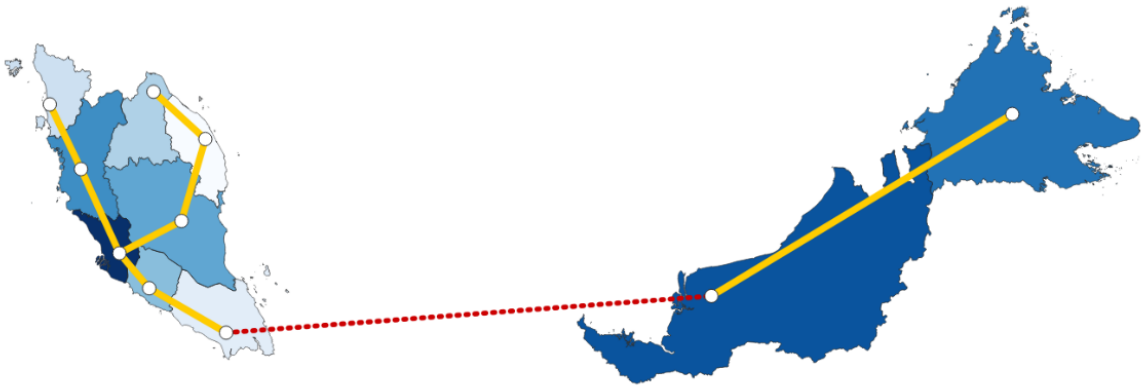
- Data/electricity.csv
- Data/assets.csv (column A values instead of column headers)
- Data/constraints.csv (column A values instead of column headers)

Interconnection identifiers should be added to the following files:

- Column A of the "Data/Calculations.xlsx" workbook to define the cost factors for each transmission line.
- Column I of the "Data/Calculations.xlsx" workbook to define the cable type and length for each transmission line.
- Column A of the "Data/factors.csv" file to add the cost factor inputs for each transmission line.

[2] Another alternative would be to use night-light raster data. The method for finding the weighted centroids for polygons superimposed over raster files in QGIS is provided in: https://spatialthoughts.com/2021/05/14/weighted-centroids-qgis-gee/

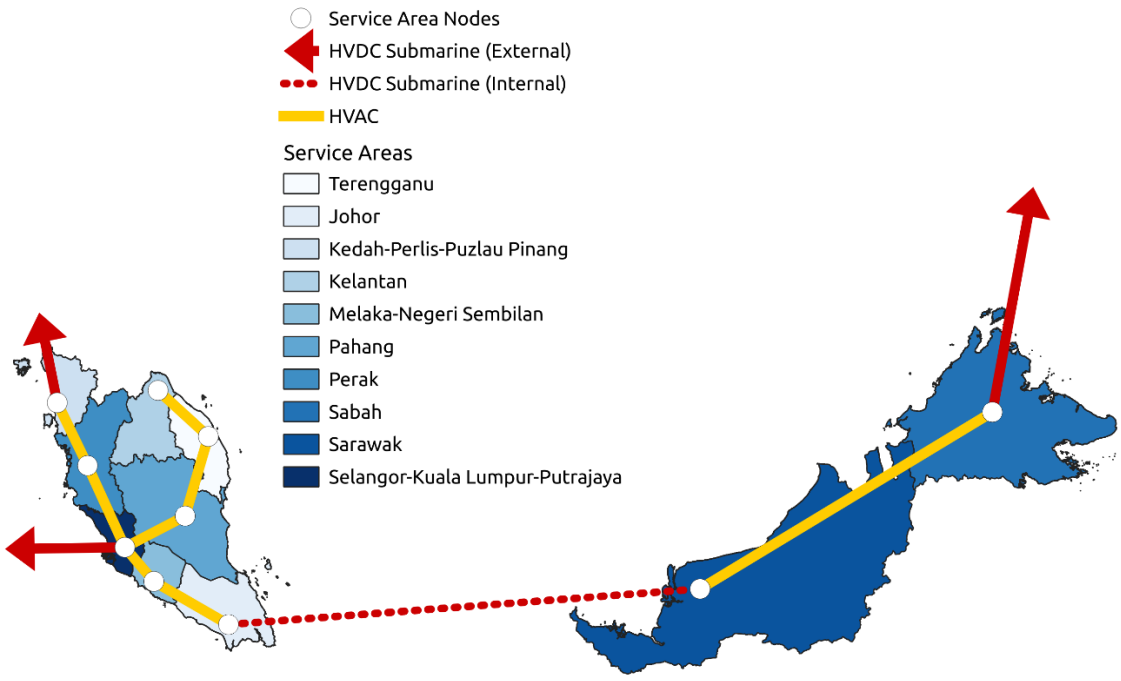
Figure 5. Electricity network for Malaysia with nodes based on population density and interconnection based on shortest transmission network



The length of each interconnection must be extracted through GIS software and added to column K of the "Data/Calculations.xlsx" workbook.

External interconnections can also be defined which allow electricity to be imported to the network from external regions. These external nodes and interconnections should also be assigned unique string identifiers. The nodal endpoints outside the network do not have any demand, solar/wind generation, storage, or hydrogen generation data applied to them. Figure 6 describes a hypothetical Malaysian electricity network with external interconnections.

Figure 6. Malaysian electricity network with external interconnections



2.2 Electricity Demand

The electricity demand inputs are stored in "Data/electricity.csv". They require two core pieces of information to develop:

- An interval electricity demand profile (e.g., 30-minute or 1-hour intervals) for a period of at least one year. If multiple years of data are available, they can be averaged together to eliminate one-off anomalies (e.g., a system-wide blackout). Note that you may choose to intentionally leave an anomaly in the data for the purposes of scenario analysis (e.g., understanding how the changes in demand due to the impact of an event like COVID-19 affects the energy storage requirements of the network).
- A scalar for the demand profile, typically based upon the population serviced at each node and an assumed demand per capita per year.

2.2.1 Demand Profile

For geographies where interval demand data is available, it is preferable to use these actual profiles for most scenarios. By using the actual demand profile for that geography, it is assumed that the hypothetical network has the same demand patterns as the existing network. That is, the same energy sub-sectors are operating in the network and are influencing demand in the same way (e.g., demand from same proportion of metal smelters or factories), and the same usage pattern of heating and cooling in offices or homes is occurring.

It is important to note that a future hypothetical electricity network will **not** actually follow the exact patterns of the current electricity network. Current electricity consumption may be dampened by the unavailability of a stable electricity supply. Industrial complexes may be using an off-grid supply in order to limit the impact of power outages to their business. Once transport has been electrified within the geography, there may be higher demand during mornings, evenings, or nights in order to charge vehicle batteries. A country that develops a large chemical industry in the future would have a demand profile modified based upon the usage pattern of this industry. This is a limitation of modelled electricity networks compared to the real world, and should be either explored through scenario analysis or expressly stated as a limitation when discussing the model outputs.

Demand profiles can generally be thought of as an average profile with some stochastic profile superimposed on top. Two common shapes for average profiles are the single-peak and twin-peak profiles (Figure 7). Single-peak profiles typically apply to warmer regions, where air-conditioning is used during the hottest parts of the day. In cooler regions, a twin-peak profile typically applies since heating is used during the cold mornings and evenings.

Average electricity demand is generally lower on weekends when offices and other workplaces are not using electricity for lighting, heating, air-conditioning, or other operations. Note that weekends may differ between countries (e.g., Nepal has a six-day workweek, one-day weekend). Similarly, public holidays will have lower demand on average. In Figure 7(a), the data for Monday is on January 26th which is the Australia Day public holiday. Again, these public holidays will be dependent upon the region that is being analysed.

Figure 7. (a) Single-peak (Australian summer week); (b) Twin-peak (Australian winter week) profiles[3]

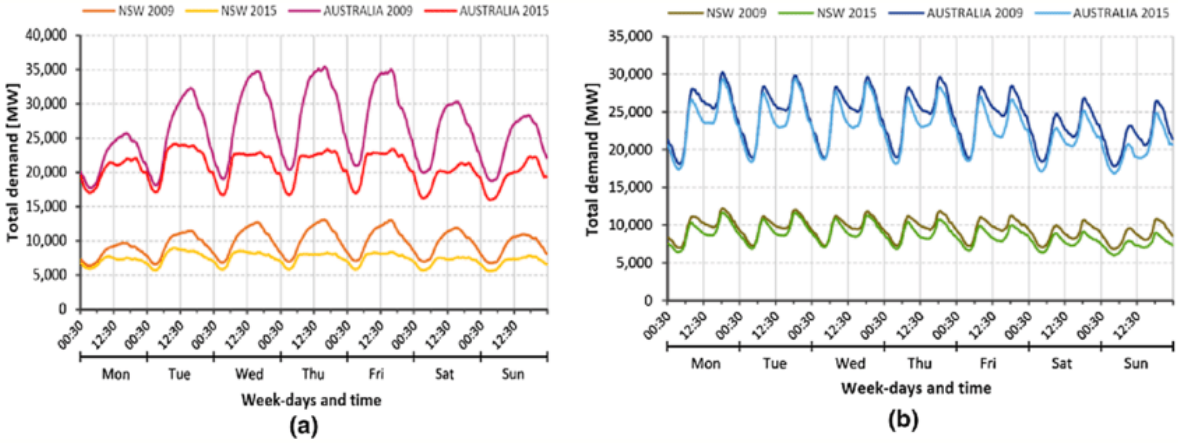
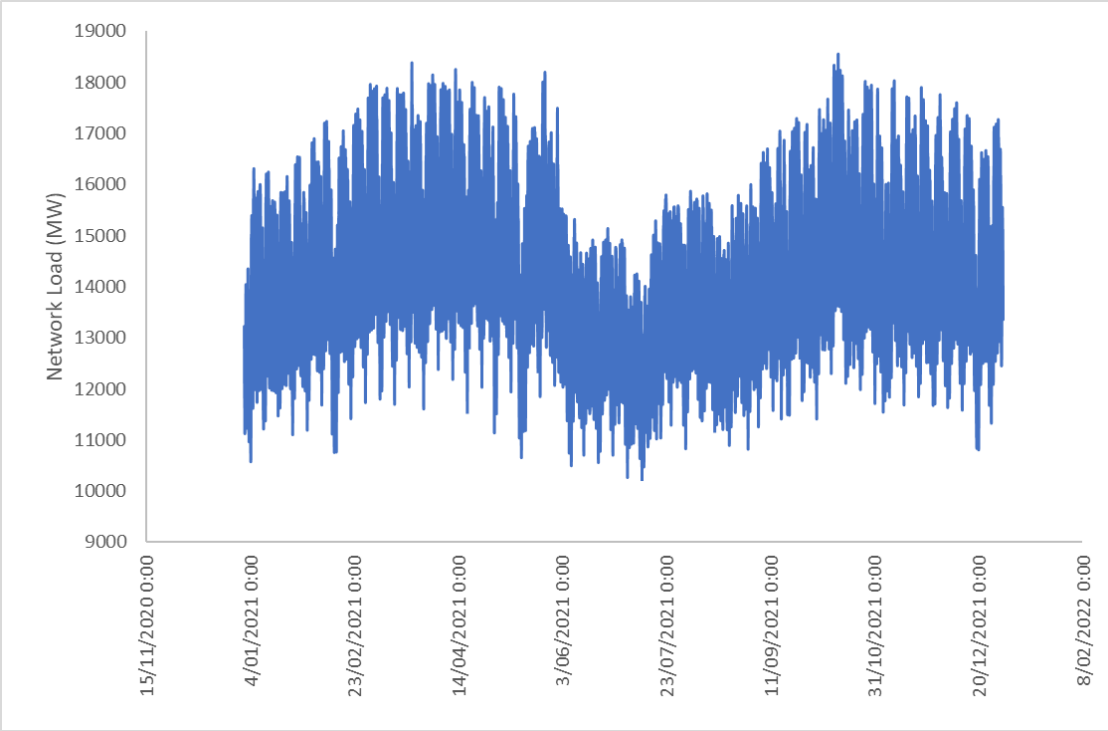
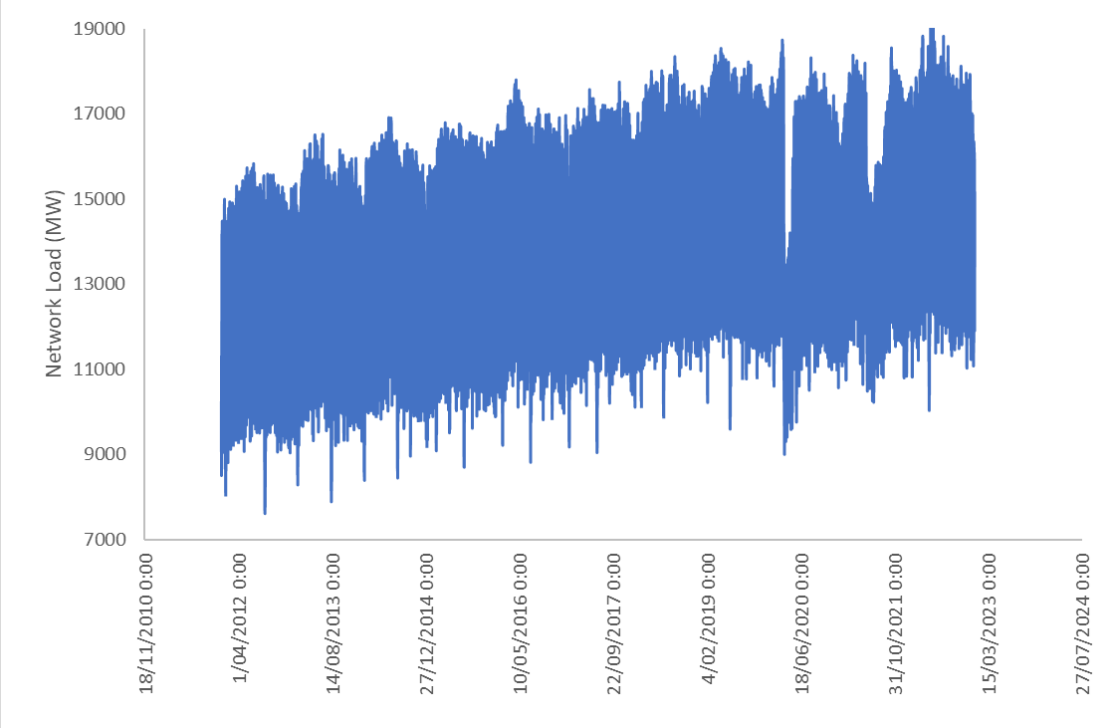


Figure 8. Peninsular grid system demand for 2021



[3] Brenna, M. et al. (2020) Numerical simulation analysis of the impact of photovoltaic systems and energy storage technologies on centralised generation: a case study for Australia. *International journal of energy and environmental engineering*. [Online] 11 (1), 9–31.

Figure 9. Peninsular grid system demand for January 2012 - December 2022

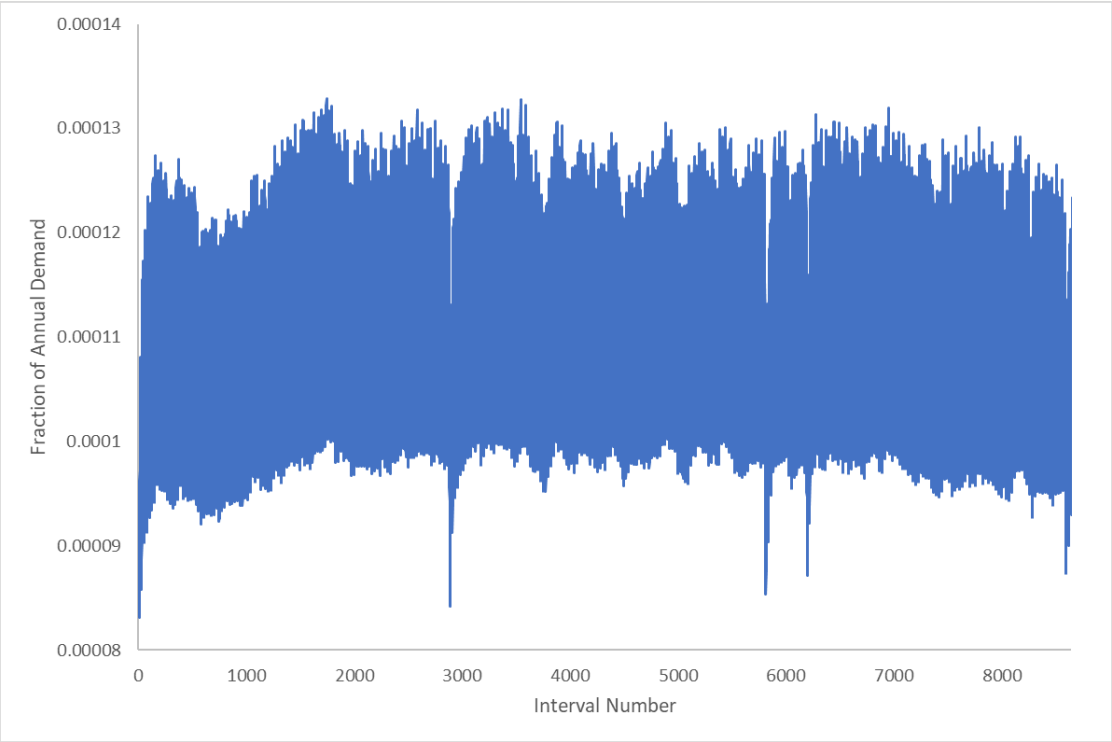


For modelling the Malaysian electricity network, the demand data has been obtained from the Grid System Operator.[4] The electricity demand for the Peninsular grid system in 2021 is shown in Figure 8. The multi-year demand for the Peninsular grid system is shown in Figure 9 to demonstrate the presence of anomalies. A severe blackout occurred in 2022, indicated by the sudden drop in demand during this period. Malaysia's Movement Control Order in response to the COVID-19 pandemic also occurred from 18 March 2020 - 3 May 2020. There is a significant drop in electricity demand during this period visible on the graph due to the temporary closure of many workplaces and industry.

If multiple years of data are available, years can be averaged together to minimise the impact of anomalies. The average demand function for a single year can then be normalised to produce the demand profile (Figure 10). To normalise the demand function, each interval should be divided by the total annual demand. In the demand profile, each interval value represents the fraction of total annual demand experienced during that interval.

[4] Grid System Operator, "System Demand Profile," 15 March 2023. [Online]. Available: https://www.gso.org.my/SystemData/SystemDemand.aspx. [Accessed 15 March 2023].

Figure 10. Peninsular grid system demand profile based on averaged yearly data from January 2012 - December 2022 (leap days removed)



Each node in the electricity network is assigned a demand profile. Within Malaysia, the grid system operators for Sarawak and Sabah do not post publicly available data. It will be assumed Sarawak and Sabah have the same demand profile as the Peninsular grid system. If nodes are in different timezones, it may be necessary to transpose demand profiles relative to the source node.

If there is no actual demand data available for a node, and if there is no demand data available for a geography that can appropriately be considered representative of that node, then a synthetic profile may be required. Techniques for developing these synthetic profiles will not be explored in detail in this documentation. One very simple profile would be to define constant demand for the entire year (this would be a safe way of finding an upper bound on storage needs, since the night-time demand would drive the need for storage while solar generation is not available). A synthetic average profile with some small stochastic values superimposed on top could also be developed for the geography, dependent upon the expected behaviour of industry and heating/cooling in that geography.

Once a demand profile has been assigned to each node in the network, they must be scaled based upon the intensity of demand at that node. Scaled demand profiles for each node are added to the "Data/electricity.csv" file. These profiles should be copied to cover the simulation period (e.g. year-long profiles should be copied 10 times to simulate over a 10-year period). Care should be taken where leap years are involved. Simply removing the intervals on 29 February from the demand profiles will avoid any issues the leap years may pose.

Demand Profile Hierarchy

The choice of demand profile is a balance of accuracy vs accessibility of data. A simple heuristic to follow is to decide on the demand profile according to the following hierarchy of choices:

1. **Actual profile for the chosen geography**
2. **Actual profile for a representative geography (e.g., a similar country)**
3. **Synthetic profile**

Ultimately, the choice of demand profile is relatively arbitrary. Modelled outputs will have very large error bars compared to the real world, regardless of the decision. It is more appropriate to analyse changes in model outputs between scenarios, rather than think of the outputs from one scenario representing a definitive reflection of reality. The need for storage is also predominantly driven by the availability of solar and wind resources, rather than minor differences between demand profile choices. The best way to understand the impact of your choice is to define a scenario for each choice, run the FIRM model and compare the results.

2.2.2 Demand Profile Scaling

The simplest way of scaling the demand profiles would be based upon a network-level demand per capita per year. Recall that the value of each interval of the demand profile represents the fraction of total annual demand at that node during the interval. The following formula should be applied to each interval of the demand profile to scale based upon a network-level demand per capita per year:

$$\begin{aligned} &Interval\ demand\ [\mathrm{MW}] \\ &\quad = Annual\ demand\ profile\ interval\ [\%] \times Service\ area\ \ population \\ &\quad \times Population\ demand\ [\mathrm{MW/person/year}] \end{aligned}$$

Instead of using a network-level population demand, a node-level population demand could be used instead. The choice depends upon the scenario analysis that is being performed.

The population of the service area can be determined from either census data (if available) or from a population raster using GIS programs.[5] If future population growth is being considered, then the following process could be used to find the service area populations:

1. Determine the current service area populations from census data or raster data.
2. Determine the fraction of total population in each service area.
3. Determine the future total population (e.g., 2050 population estimated by the World Bank[6]).
4. Multiply the fraction of total population in each service area by the future total population estimate.

For the Selangor-Kuala Lumpur-Putrajaya node, the Malaysia demand profile has been scaled by 10 MWh per person per year (Figure 11). This defines a scenario where Malaysia's electricity demand has caught up to the demand of Singapore. The total population of Malaysia in 2050 was estimated to be 41,032,433 in accordance with the World Bank estimate. High resolution population density maps developed by Meta were used to estimate the proportion of total population in each service area using QGIS.[7]

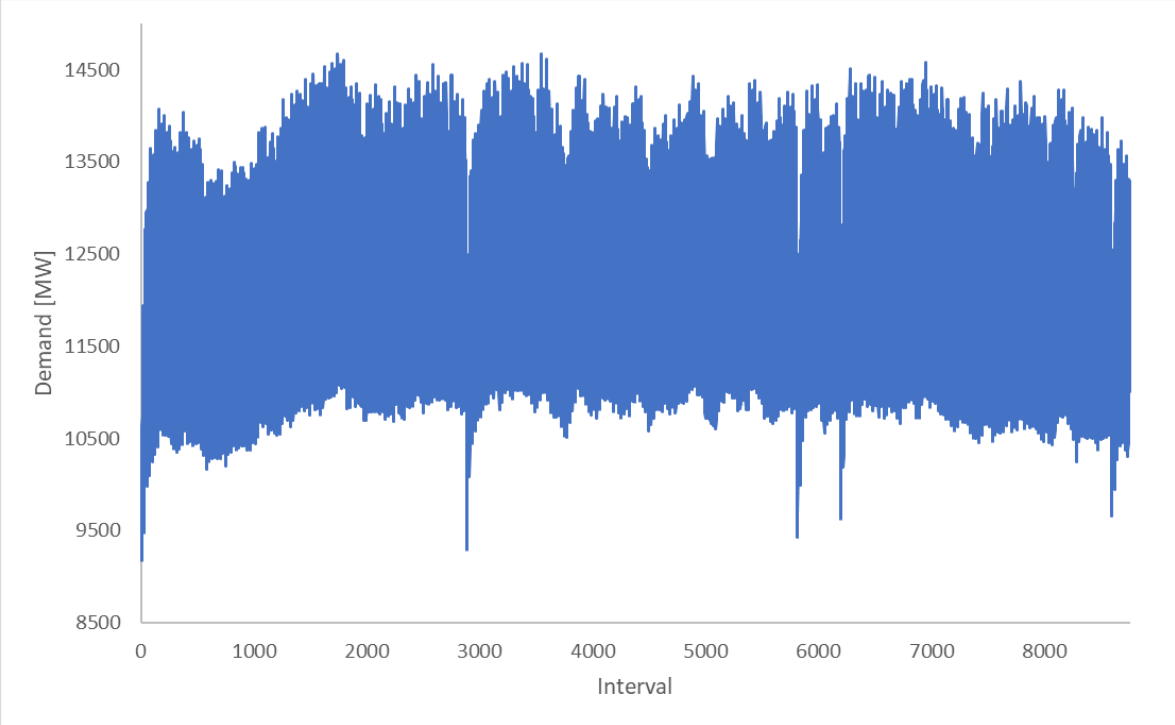
An example "Data/electricity.csv" file for Malaysia is shown in Figure 12.

[5] The process for summing raster values (e.g., population raster data) within a vector polygon (e.g. service area polygons) in QGIS is described here: https://docs.qgis.org/2.18/en/docs/user_manual/plugins/plugins_zonal_statistics.html

[6] World Bank Database: https://databank.worldbank.org/source/population-estimates-and-projections

[7] Data for Good at Meta, "Malaysia: High Resolution Population Density Maps + Demographic Estimates", 18 February 2023. [Online]. Available at: https://data.humdata.org/dataset/malaysia-high-resolution-population-density-maps-demographic-estimates. [Accessed 4 April 2023].

Figure 11. Selangor-Kuala Lumpur-Putrajaya demand function assuming 10 MWh per person per year scaling



This might be per region electricity demand per hour in MW, after applying the scaling formula

Figure 12. Example demand data for Malaysia FIRM from "Data/electricity.csv"

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Year	Month	Day	Hour	ME	SB	TE	PA	SE	PE	JO	KT	KD	SW	TH	IN	PH
2	2013	1	1	1	1279.181	2382.16	666.0581	966.9146	5293.907	1546.091	2364.262	961.8571	2554.561	1642.108	0	0	0
3	2013	1	1	2	1237.006	2303.619	644.0979	935.0351	5119.365	1495.116	2286.312	930.1443	2470.336	1587.967	0	0	0
4	2013	1	1	3	1202.189	2238.782	625.9693	908.7179	4975.277	1453.035	2221.962	903.9648	2400.807	1543.272	0	0	0
5	2013	1	1	4	1180.443	2198.285	614.6461	892.28	4885.279	1426.751	2181.769	887.6129	2357.379	1515.356	0	0	0
6	2013	1	1	5	1175.271	2188.653	611.9531	888.3705	4863.874	1420.5	2172.209	883.7238	2347.05	1508.716	0	0	0
7	2013	1	1	6	1158.057	2156.597	602.9902	875.3591	4792.636	1399.694	2140.394	870.7804	2312.674	1486.619	0	0	0
8	2013	1	1	7	1109.392	2065.97	577.6507	838.5738	4591.235	1340.875	2050.448	834.1876	2215.488	1424.147	0	0	0
9	2013	1	1	8	1157.818	2156.151	602.8654	875.1779	4791.644	1399.405	2139.951	870.6003	2312.196	1486.312	0	0	0
10	2013	1	1	9	1222.099	2275.858	636.336	923.7671	5057.672	1477.099	2258.76	918.9352	2440.567	1568.83	0	0	0
11	2013	1	1	10	1269.476	2364.087	661.0048	959.5787	5253.743	1534.361	2346.325	954.5596	2535.18	1629.649	0	0	0
12	2013	1	1	11	1280.519	2384.652	666.7548	967.926	5299.444	1547.708	2366.735	962.8632	2557.233	1643.825	0	0	0
13	2013	1	1	12	1270.704	2366.374	661.6443	960.507	5258.825	1535.846	2348.595	955.483	2537.633	1631.226	0	0	0
14	2013	1	1	13	1285.691	2394.283	669.4478	971.8354	5320.849	1553.96	2376.295	966.7522	2567.562	1650.465	0	0	0
15	2013	1	1	14	1299.669	2420.315	676.7263	982.4016	5378.699	1570.855	2402.131	977.2631	2595.477	1668.409	0	0	0
16	2013	1	1	15	1306.748	2433.498	680.4123	987.7525	5407.996	1579.411	2415.215	982.5861	2609.615	1677.497	0	0	0
17	2013	1	1	16	1300.818	2422.453	677.3242	983.2695	5383.451	1572.243	2404.253	978.1264	2597.77	1669.883	0	0	0
18	2013	1	1	17	1285.861	2394.599	669.5362	971.9637	5321.551	1554.165	2376.608	966.8798	2567.901	1650.683	0	0	0
19	2013	1	1	18	1329.883	2476.58	692.4582	1005.24	5503.738	1607.373	2457.973	999.9815	2655.815	1707.195	0	0	0
20	2013	1	1	19	1440.013	2681.671	749.8021	1088.486	5959.514	1740.483	2661.523	1082.792	2875.748	1848.571	0	0	0
21	2013	1	1	20	1442.939	2687.119	751.3254	1090.697	5971.621	1744.018	2666.93	1084.992	2881.591	1852.327	0	0	0
22	2013	1	1	21	1414.702	2634.535	736.6229	1069.353	5854.764	1709.89	2614.742	1063.76	2825.201	1816.079	0	0	0
23	2013	1	1	22	1382.162	2573.938	719.6797	1044.757	5720.097	1670.561	2554.599	1039.292	2760.218	1774.307	0	0	0
24	2013	1	1	23	1329.074	2475.074	692.0371	1004.628	5500.391	1606.395	2456.478	999.3734	2654.199	1706.157	0	0	0
25	2013	1	1	24	1369.502	2550.361	713.0874	1035.187	5667.702	1655.258	2531.199	1029.772	2734.935	1758.054	0	0	0
26	2013	1	2	1	1217.885	2268.012	634.142	920.5821	5040.234	1472.006	2250.972	915.767	2432.152	1563.421	0	0	0

2.3 Solar

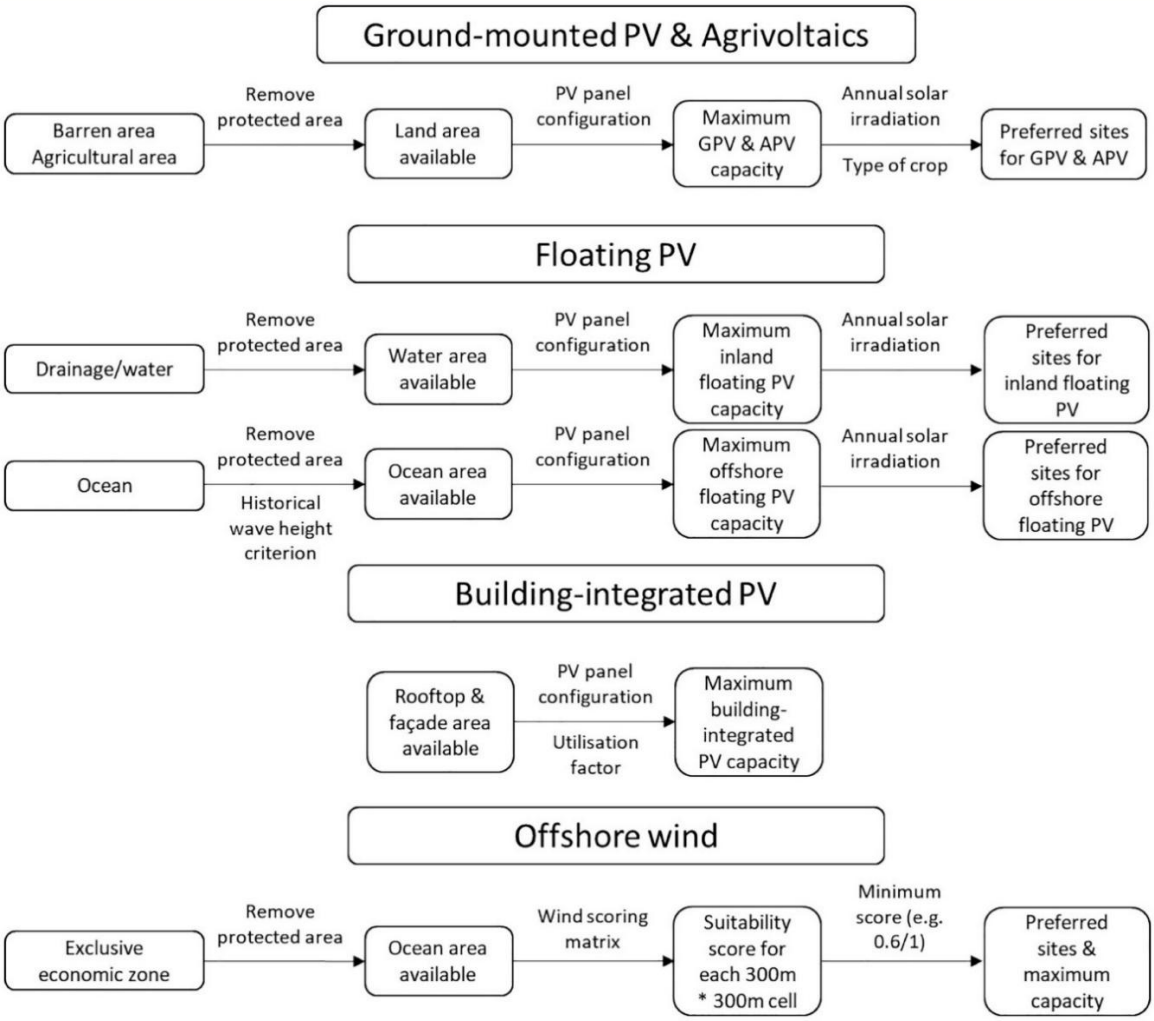
Solar resources are defined according to solar profiles that indicate the average capacity factor at each interval for solar sites within each service area. The first step in developing these solar profiles is to define locations for solar sites in each service area.

The simplest method of choosing the locations of each solar site is to simply define one solar site at the location of each node. This method does not require a detailed solar resource analysis, but comes with inherent risks. Nodes are likely not to be the optimal location for solar sites to be constructed, so it is likely that the solar capacity in the least-cost optimisation would be an overestimate. Furthermore, only choosing one solar site location per node opens them up to the risk of anomalous events influencing the results (e.g., extended cloudy

periods or reduced solar insolation due to screening from bushfire smoke). The need for storage is heavily driven by these anomalous events, so an unreasonably large storage capacity may be calculated by the FIRM model.

A solar and offshore wind site selection process using GIS is described in Figure 13. The process also allows resource constraints for solar and wind at each site to be defined within the FIRM model. Further details of the process are provided in Cheng et al. (2022).[8]

Figure 13. GIS analysis process for the resource assessment of solar PV and offshore wind



Once the solar site locations have been identified, interval data for solar insolation over the simulation period must be obtained. Solcast provides solar insolation data based upon satellite data. The solar insolation for a 16-year period at the SW2 solar site (1.671°N, 111.257°E) is provided in Figure 14. Care must be taken where leap years are involved. The intervals in the solar profiles should match the intervals in the demand profiles.

[8] C. Cheng, A. Blakers, M. Stocks and B. Lu, "100% Renewable Energy in Japan," *Energy Conversion and Management,* vol. 255, no. 115299, 2022.

Figure 14. Solar insolation for SW2 from January 2013 - December 2022 (leap days removed)

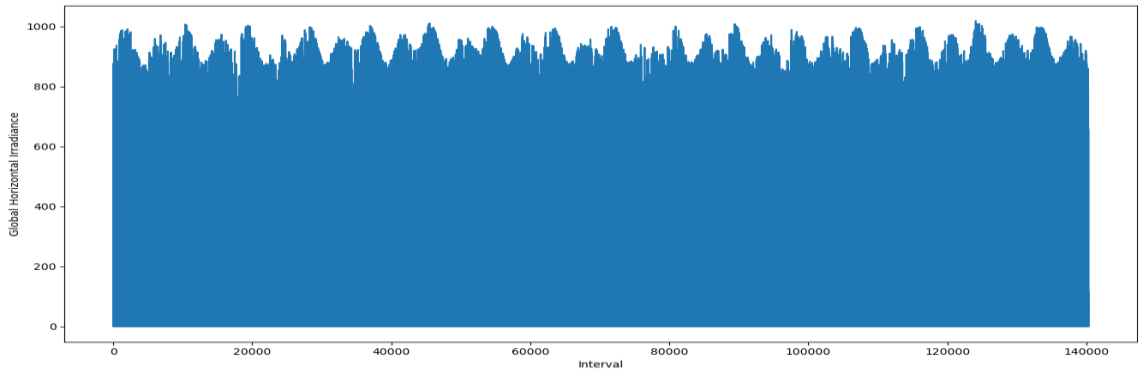
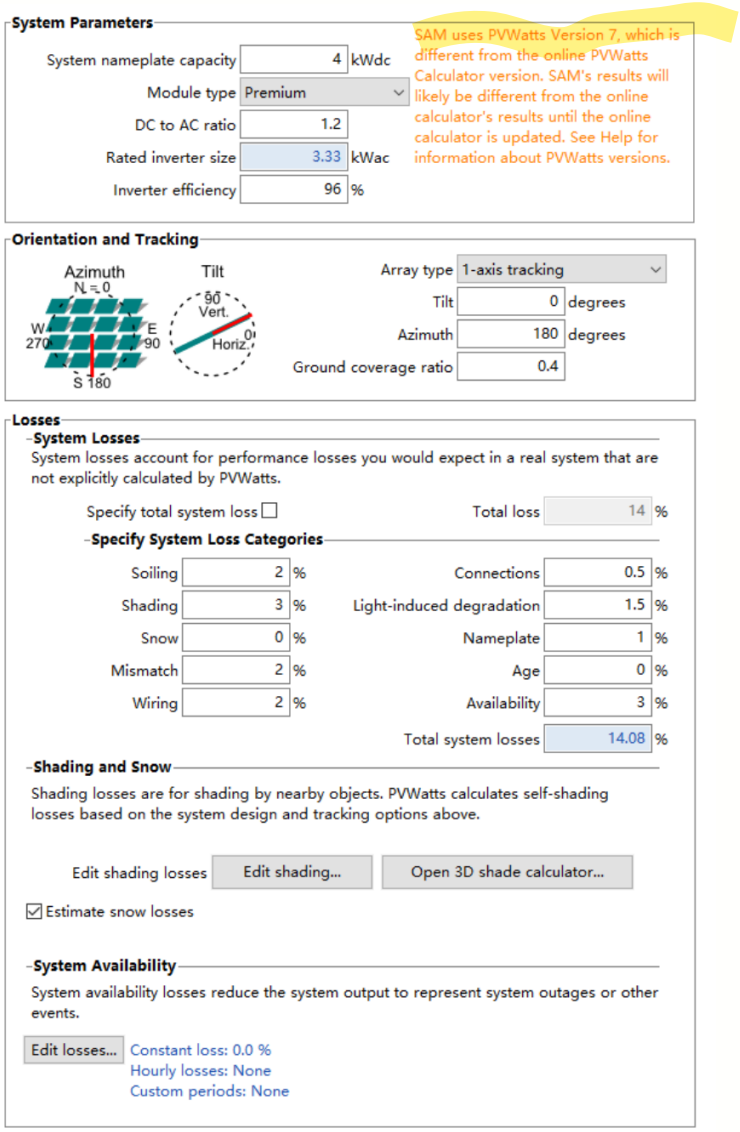
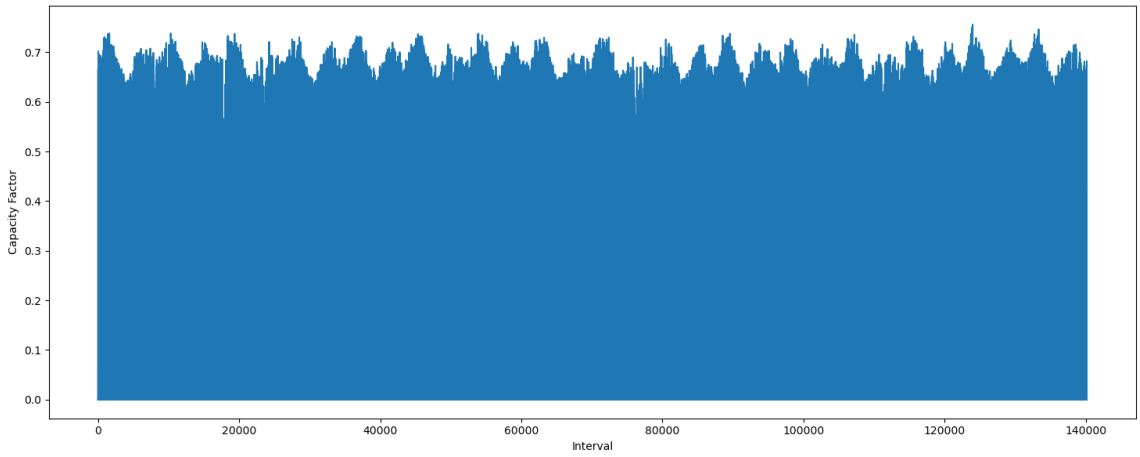


Figure 15. Example System Advisor Model inputs for solar site



This solar insolation data can then be run through the System Advisor Model[9] to obtain the solar profiles for each solar site. The data output by the System Advisor Model represents the power output of a solar site receiving the input solar insolation for each interval. The solar profile can be obtained by dividing the power output at each interval by the capacity of the solar site defined by the System Advisor Model. An example set of inputs for the System Advisor Model are provided in Figure 15. The 16-year solar profile for the SW2 solar site is shown in Figure 16.

Figure 16. Solar profile for SW2 from January 2013 - December 2022 (leap days removed)



For a large number of solar sites, it may be more efficient to automate the generation of solar profiles using the PySAM.Pvwattsv8 package for Python. An example script is available on Github[10] which converts a Solcast solar insolation file to an NREL solar insolation file. A set of NREL solar insolation files can then be processed by the System Advisor Model using Pvwattsv8 according to another example script[11].

Each solar site should be assigned a unique string identifier. These identifiers should be added to "Data/pv.csv" as the column headers. The solar profile for the corresponding solar site should be added under each header. Solar identifiers should be added in the order of nodes contained in the "Data/electricity.csv" file (solar sites connected to the same node should be consecutive). A screenshot of the "Data/pv.csv" file including the Malaysia solar profiles is shown in Figure 17.

[9] The System Advisor Model is free to download from the National Renewable Energy Laboratory website: https://sam.nrel.gov/download.html

[10] https://github.com/TimWeberRE100/FIRM_Malaysia_Complex/blob/main/Tools/solcast_to_nrel.py

[11] https://github.com/TimWeberRE100/FIRM_Malaysia_Complex/blob/main/Tools/pvwatts_sdk.py

Figure 17. Malaysia FIRM model input file for solar profiles

	A	B	C	D	E	F	G	H	I	J	K	L
1	Year	Month	Day	Hour	ME	SB1	SB2	TE	PA	SE	PE1	PE2
2	2007	1	1	1	0	0	0	0	0	0	0	
3	2007	1	1	2	0	0	0	0	0	0	0	
4	2007	1	1	3	0	0	0	0	0	0	0	
5	2007	1	1	4	0	0	0	0	0	0	0	
6	2007	1	1	5	0	0	0	0	0	0	0	
7	2007	1	1	6	0	0	0	0	0	0	0	
8	2007	1	1	7	0	0	0	0	0	0	0	
9	2007	1	1	8	0	0.0429801635692761	0.174699233727596	0	0	0	0	
10	2007	1	1	9	0.0636251361853884	0.242260306709929	0.446893986924116	0.130580548651018	0.165271473272803	0.0974714624284801	0.185559565100487	0.127983
11	2007	1	1	10	0.16540349117068	0.517030720274717	0.567726673550506	0.132107202110283	0.212123691928581	0.320187145550258	0.465588543451936	0.313709
12	2007	1	1	11	0.205420352647406	0.536057060489475	0.325099662673655	0.12803461123374	0.554670637921107	0.199987626505713	0.43496366075172	0.588916
13	2007	1	1	12	0.505882362788164	0.178059153140581	0.403564812662294	0.167358652584389	0.650177959747442	0.274575404545521	0.509348628788025	0.623668
14	2007	1	1	13	0.567769290923905	0.164916155297017	0.469606819445875	0.171385474498488	0.413094802796664	0.278352054247702	0.555326317144583	0.647011
15	2007	1	1	14	0.259214919890044	0.514828477833035	0.417708100648679	0.205134648958967	0.587624141650777	0.382261885122441	0.422016834155957	0.577011
16	2007	1	1	15	0.163377569622037	0.445111868089917	0.461207931924279	0.196621163917651	0.630125874949758	0.310192798901856	0.409022689724634	0.523480
17	2007	1	1	16	0.134437768867645	0.115514268225767	0.574945595046622	0.096110276093339 3	0.522719986109992	0.20722470836966	0.354987126609759	0.386563
18	2007	1	1	17	0.104449313373088	0.06303067687581	0.316183586558392	0.0630594999125459	0.250358569959037	0.0869601787461838	0.118038103319751	0.0601728
19	2007	1	1	18	0.0437382746965247	0.0225307656477726	0.03631546526604	0.0495711789574578	0.0658263670761341	0.0257560488335753	0.0343220187749795	0.0330943
20	2007	1	1	19	0.0116121382923924	0	0	0.0145241109199028	0.0173332714948222	0.0116579173978405	0.00658953414780107	0.0104716
21	2007	1	1	20	0	0	0	0	0	0	0	
22	2007	1	1	21	0	0	0	0	0	0	0	
23	2007	1	1	22	0	0	0	0	0	0	0	
24	2007	1	1	23	0	0	0	0	0	0	0	
25	2007	1	1	24	0	0	0	0	0	0	0	
26	2007	1	2	1	0	0	0	0	0	0	0	
27	2007	1	2	2	0	0	0	0	0	0	0	
28	2007	1	2	3	0	0	0	0	0	0	0	
29	2007	1	2	4	0	0	0	0	0	0	0	
30	2007	1	2	5	0	0	0	0	0	0	0	
31	2007	1	2	6	0	0	0	0	0	0	0	
32	2007	1	2	7	0	0	0	0	0	0	0	
33	2007	1	2	8	0	0.0721848051172957	0.162205426245603	0	0	0	0	
34	2007	1	2	9	0.215704991558259	0.129745557682324	0.573033624967674	0.126193470606345	0.208865702776654	0.11696258622859	0.171242254788842	0.0976870
35	2007	1	2	10	0.482529353255602	0.265115156228823	0.631167206702897	0.138162743872863	0.393301533240134	0.321179399371782	0.371640948044479	0.244066
36	2007	1	2	11	0.587223981286592	0.37766889058783	0.60661634209754	0.529843797143819	0.559309208602151	0.363648954848667	0.501328292950609	0.466147
37	2007	1	2	12	0.617966949672164	0.430676084327372	0.661673135333971	0.66793641358106	0.612830567667691	0.520129436472877	0.62657447683967	0.604384
38	2007	1	2	13	0.617367467754326	0.609996123019747	0.610125998792294	0.670096444640472	0.627066748167025	0.373457287197618	0.646750077438959	0.643922

2.4 Wind

Similar to the solar inputs, wind resources are defined according to wind profiles that indicate the average capacity factor at each interval for wind sites within each service area.

Wind site locations can also be simply defined by assigning one wind site to each nodal location. Such a method carries the same risk of anomalous events driving the FIRM model outputs if there are periods of unusually low wind speeds.

Offshore wind sites could also be selected based on GIS analysis following the same process described in Figure 13. The process could be modified for onshore wind based upon the available land for wind farms.

Wind speed data can be obtained from local weather stations, Solcast, or ERA5 Reanalysis and input into the System Advisor Model.

In a similar way to the solar data, the System Advisor Model will produce the hourly power output for a given wind system which can be used to calculate the hourly capacity factor for that wind system. Alternatively, the ANU's global wind atlas can be used to obtain hourly wind capacity factor profiles for offshore wind sites.[12]

The wind sites are each assigned a unique string identifier, which are used as the column headers in "Data/wind.csv". Wind profiles can then be copied below the corresponding identifier within the input file. Wind identifiers should be added in the order of nodes contained in the "Data/electricity.csv" file (wind sites connected to the same node should be consecutive).

2.5 Hydrogen

Hydrogen is used during peak periods to reduce the need for seasonal storage. It is considered to be the most expensive flexible generation technology within the FIRM model and is only dispatched when energy deficits cannot be met by all other baseload and flexible generation technologies.

Constraints on the maximum capacity (GW) and maximum annual generation (GWh) from hydrogen at each node should be defined in the "Data/assets.csv" and "Data/constraints.csv" files. This will limit the size of the solution space during the least-cost optimisation and ensure that an unreasonable amount of hydrogen is not dispatched for firming over the simulation period.

[12] Available from: http://windatlas.xyz/

2.6 Existing Capacity

As specified in section 1.1, solar PV, wind, and hydrogen are assumed to be the only new source of generation capacity within the FIRM model. Existing capacity for biomass, hydroelectric, thermal power stations, and nuclear can still be included as either baseload or flexible generation.

Data sources for existing capacity will be dependent upon the geography covered by the electricity network.

Existing capacity for each technology at each node is added to the "Data/assets.csv" file. Capacity is given in units of MW. The FIRM model will assume that all existing capacity is for flexible generation (e.g., hydroelectric is storage hydro). Load-following nuclear capacity or load-following thermal should be considered flexible capacity.

A fraction of the generation from each technology can instead be defined to be baseload generation within the "Inputs.py" module (see section 3.1). For hydro, this fraction should be based upon the proportion of hydroelectric capacity that is run-of-river. Baseload generation is automatically dispatched first to meet demand for each interval during the least-cost optimisation.

it is assets_existing

An alternative method of defining baseload generation is to create a new input file called "Data/baseload.csv". The file would have the same structure as the "Data/electricity.csv" file except instead of specifying interval demand for each node, it would define interval baseload generation at each node. This allows baseload generation to be defined as a function of time, which is essential for capturing the seasonal behaviour of run-of-river hydroelectric systems.

An example of the existing capacity input file is provided in Figure 18.

Figure 18. Example of "Data/assets.csv" file for Malaysia

	Node	Code	Fraction	Hydro	Biomass
2	Melaka	ME	1	0	0
3	Sabah	SB	1	250	75
4	Terengganu	TE	1	670	0
5	Pahang	PA	1	930	75
6	Selangor	SE	1	0	75
7	Perak	PE	1	660	75
8	Johor	JO	1	0	75
9	Kelantan	KT	1	610	75
10	Kedeh	KD	1	0	0
11	Sarawak	SW	1	4790	0
12	Thailand	TH	1	0	0
13	Indonesia	IN	1	0	0
14	Philippines	PH	1	0	0

Annual generation from each flexible technology can be constrained using the "Data/constraints.csv" input file. The constraint at each node is defined in units of GWh. Annual generation from each technology cannot exceed the constraint for that corresponding node. An example "Data/constraints.csv" file is given in Figure 19.

Figure 19. Example of "Data/constraints.csv" for Malaysia

1	Node	Code	Fraction	Hydro	Biomass
2	Melaka	ME	1	0	0
3	Sabah	SB	1	460	130
4	Terengganu	TE	1	2200	0
5	Pahang	PA	1	2000	40
6	Selangor	SE	1	0	40
7	Perak	PE	1	1500	40
8	Johor	JO	1	0	40
9	Kelantan	KT	1	1400	40
10	Kedeh	KD	1	0	0
11	Sarawak	SW	1	21000	0
12	Thailand	TH	1	0	0
13	Indonesia	IN	1	0	0
14	Philippines	PH	1	0	0

2.7 Cost Factors

Cost factors are assigned to each generation technology, interconnection, and storage technology. The capital cost (annualised $/W), fixed operation and maintenance cost ($/W-year), and variable operation and maintenance costs ($/Wh) each have a corresponding cost factor. Cost factors for the same technology and unit (power or energy) are then summed together.

Cost factors account for the real discount rate, with the `PV` Excel function (present value) used to determine the present value of the capital costs over the lifetime of the system.

Cost factor calculations are performed in the "Data/Calculations.xlsx" workbook. The key for each default cost factor identifier is provided in Table 1.

Final cost factors are calculated on the "factor" sheet of the "Data/Calcuations.xlsx" workbook. They must be copied into the "Data/factors.csv" file.

Table 1. Key for default cost factor identifiers

Identifier	Meaning	Cost Factor Units
PV	Solar photovoltaic capital cost	$/W-year
PV-FOM	Solar photovoltaic fixed operation and maintenance cost	$/W-year
Wind	Wind capital cost	$/W-year
Wind-FOM	Wind fixed operation and maintenance cost	$/W-year
Inter	Cost of electricity imported from external regions	$/kWh
PHP	Pumped hydro power capacity capital cost	$/W-year
PHES-FOM	Pumped hydro fixed operation and maintenance cost	$/W-year
PHES-POM	Pumped hydro periodic operation and maintenance (e.g., refurbishing after 40 years)	$/W-year
PHS	Pumped hydro storage capacity capital cost	$/Wh-year
BP	Battery power capacity capital cost	$/W-year
BESS-FOM	Battery fixed operation and maintenance cost	$/W-year
BESS-POM	Battery periodic operation and maintenance (e.g., refurbishing after 10 years)	$/W-year
BS	Battery storage capacity capital cost	$/Wh-year
Transmission Identifier	Capital cost for a specific interconnection	$/W-year

Identifier	Meaning	Cost Factor Units
Transmission Identifier FOM	Fixed operation and maintenance cost for a specific interconnection	$/W-year
Transmission Identifier Converter	Capital cost for the HVAC to HVDC converter of a specific interconnection	$/W-year
Transmission Identifier Converter-FOM	Fixed operation and maintenance cost for the HVAC to HVDC converter of a specific interconnection	$/W-year
ACPV	Capital cost for the AC connection of a photovoltaic site	$/W-year
ACWind	Capital cost for the AC interconnection of a wind site	$/W-year
Hydro	Levelised cost of electricity for existing hydro	$/kWh
Bio	Levelised cost of electricity for existing bioenergy	$/kWh
Gas	Capital cost for hydrogen fuel cell	$/W-year
Gas-FOM	Fixed operation and maintenance cost for hydrogen fuel cell	$/W-year
Gas-VOM	Variable operation and maintenance cost for hydrogen fuel cell	$/Wh
Gas-Fuel	Cost of hydrogen fuel required to generate electricity	$/Wh
PHES-VOM	Pumped hydro variable operation and maintenance cost	$/Wh discharged
LegPH	Annual cost of legacy pumped hydro	$b/year

Identifier	Meaning	Cost Factor Units
LegINTC	Annual cost of legacy interconnections	$b/year
Fossil	Levelised cost of electricity for existing fossil fuel power stations	$/kWh

3 Modules

Running the default "Optimisation.py" script will run the entire FIRM model simulation and generate outputs. Each of the other modules are imported at different points in the simulation. Command line arguments are used when running the "Optimisation.py" script to change optimisation parameters or select scenarios for analysis. An example FIRM model for Malaysia is provided on Github and should be referred to when reading through this section.[13]

The code for each module in the FIRM model must be tailored to suit the particular hypothetical electricity network and assumed generation, storage, and transmission technologies. It is suitable to start with an existing codebase for a FIRM model, then modify it to suit the particular set of scenarios for a new system.

3.1 Inputs

The input module imports all of the data in the "Data" folder and establishes all of the assumptions. The time series demand data, energy resource data, energy constraints and resource constraints are used to define variables in this module.

3.1.1 Imports

The `numpy` package is imported to enable the use of numpy arrays and handle the CSV files containing raw data.

The `scenario`, `node`, and `percapita` variables are imported from the "Optimisation.py" module. These variables are defined by command line arguments and are used to select between scenarios that have different electricity network shapes or data assumptions.

3.1.2 Nodal Lists Network.py

List variables are defined for the node unique identifiers, solar sites for each node, wind sites for each node, and external interconnections for each node. The node multipliers in the `PVl`, `Windl`, and `Interl` lists indicate the number of solar sites, wind sites, and external interconnections at each node respectively. The `resolution` variable defines the length of each interval, with a value of 1 indicating intervals that are 1 hour long (e.g., 30-minute intervals would have a resolution of 0.5).

The `pv_ub_np`, `phes_ub_np`, and `wind_ub_np` variables can be used to define upper bound constraints on the capacity of solar PV, PHES, and wind at respective sites. Values are given in units of GW.

Input.py

3.1.3 Data Imports

The resource data, electricity demand, and existing capacities are used to define the following variables:

- `MLoad`: The interval electricity demand is imported from "Data/electricity.csv"
- `TSPV`: The solar profiles are imported from "Data/pv.csv"
- `TSWind`: The wind profiles are imported from "Data/wind.csv"
- `CFlexible`: The existing capacities for each technology are imported from "Data/assets.csv"
- `EFlexible`: The constraints on generation from existing capacity are imported from "Data/constraints.csv"
- `CBaseload`: Baseload capacity for a single time interval is defined according to the fraction of each existing capacity technology assigned to baseload generation. It is based on the annual generation constraints (in GWh) divided by the number of intervals in each year (this accounts for the annual capacity factor of that technology).

[13] https://github.com/TimWeberRE100/FIRM_Malaysia_Complex

- `CPeak`: Peaking capacity for a single time interval equals the sum of all flexible technology capacities minus the baseload capacity from each of those technologies. Baseload capacity is based on the annual generation constraints (in GWh) divided by the number of intervals in the year.
- `baseload`: Generation profile for baseload capacity over the entire simulation period. Can either be defined by `CBaseload` or by "Data/baseload.csv" if time-series generation profiles are preferred.

3.1.4 Constraints

Energy and transmission constraints are defined according to the following variables:

- *`Existing`*`max`: Constraints on annual energy generated by a particular existing capacity technology.
- `externalImports`: Constraint that restricts the amount of electricity that can be imported from external interconnections. A value of 0.05 indicates that 5% of demand can be balanced by external interconnections.
- `CDC`*`n`*`max`: The maximum capacity of specific interconnections, constraining electricity flows along those interconnections.

3.1.5 Transmission Losses

Input.py

Transmission losses for each interconnection will depend upon whether that line is high-voltage AC or high-voltage DC. The following variables are used to specify transmission losses for each interconnection:

- `dc_flags`: An array of Boolean values that define whether or not an interconnection is a high-voltage DC line.
- `TDistances`: The length, in kilometres, of each interconnection. Interconnections in the list must have the same order as the `dc_flags` array.
- `TLoss`: The electricity losses along the length of each interconnection. Typically, 3% per 1,000 km for HVDC and 7% per 1,000 km for HVAC.[14]

3.1.6 Storage System Constants, Cost Factors and Simulation Period

Additional data required for the FIRM simulation:

- `efficiency`*`SS`*: Round-trip efficiency of the storage system. Default value of 80% for pumped hydro energy storage.
- `factor`: The cost factors are imported from "Data/factor.csv"
- `firstyear, finalyear, timestep`: The years covering the simulation period (inclusive) and the size of the time intervals. A `timestep` of 1 indicates intervals that are 1 hour long.

3.1.7 Scenario Adjustments

The `node` and `scenario` variables imported from "Optimisation.py" are used to select different network design scenarios. In this block, `if` statements based upon the value of `node` are used to modify the electricity network lists, data imports and constraints based upon the alternative network designs.

Defining `node` to be the unique identifier of a specific node will change the values of all other nodes to 0 for these variables. That is, the scenario will analyse that node with no interconnection to any other node.

The `coverage` variable contains an array of all nodes covered by the specific scenario. The network lists, data imports and constraints that were previously defined for the entire network, can be filtered to only include data based upon the node coverage.

[14] https://iea-etsap.org/E-TechDS/PDF/E12_el-t&d_KV_Apr2014_GSOK.pdf

3.1.8 Decision Variable List Indexes Input.py

The decision variables are stored in a list. The following indexes are used to locate decision variables in that list:

- `intervals, nodes`: The number of intervals equals the number of rows in the `MLoad` variable. The number of nodes equals the number of columns in `MLoad`.
- `years`: The number of years in the simulation period.
- `pzones, wzones`: Number of solar sites and wind sites.
- `pidx, widx, phidx, bidx`: Integers that define the final index of solar sites, wind sites, pumped hydro sites and battery sites within the decision variable list.
- `inters`: The number of external interconnections.
- `iidx`: Integer that defines the final index of capacity imported from external nodes in the decision variable list. Note that the pumped hydro and battery energy capacity is stored between the battery sites and external interconnections, which is the reason that 2 is added to the sum.
- `gidx`: Integer that defines the final index of gas sites in the decision variable list.

3.1.9 Network Constraints

Additional constraints on allowable deficit and storage power capacity:

- `energy`: The total annual electricity demand for the entire network.
- `contingency_ph`: A list that defines the lower bound of pumped hydro power capacity at each node.
- `contingency_b`: A list that defines the lower bound of battery power capacity at each node.
- `manage`: The number of weeks during which load shedding is allowed to occur.
- `allowance`: The allowable deficit energy over the simulation period. Indicates high-stress periods where load shedding may occur.
- `Gasmax`: The maximum amount of annual electricity demand that can be balanced using hydrogen within the simulation.

3.1.10 Decision Variable Upper Bounds

Lists of upper bound values defined for all decision variables optimised through the differential evolution:

- `pv_ub`: Upper bounds on solar PV capacity at each solar site. Units of GW.
- `wind_ub`: Upper bounds on wind capacity at each wind site. Units of GW.
- `phes_ub`: Upper bounds on pumped hydro power capacity at each node. Units of GW.
- `battery_ub`: Upper bounds on battery power capacity at each node. Units of GW.
- `phes_s_ub`: Upper bounds on pumped hydro energy capacity at each node. Units of GWh.
- `battery_s_ub`: Upper bounds on battery energy capacity at each node. Units of GWh.
- `inter_ub`: Upper bounds on external interconnection capacity at each external node. Units of GW.
- `gas_ub`: Upper bounds on hydrogen capacity at each node. Units of GW.

3.1.11 Solution Class

A class that defines objects that store the candidate solutions of decision variables and other solution-related variables. All decision variables are stored in the `x` variable. This includes the total capacity of and generation power from solar photovoltaics at each solar site, total capacity of and generation power from wind at each wind site, power and energy capacity of pumped hydro, power and energy capacity of batteries, capacity and energy flows of external interconnections, and capacity of hydrogen.

The solution object also stores non-decision variables that were defined in the module, including existing capacity, allowance, electricity network lists, efficiency, baseload, resolution, electricity demand array, number of intervals, and number of nodes. These variables are required by other modules for the definition of the electricity network and calculation of output results.

3.2 Simulation

For a given solution object, energy balancing is performed assuming that there is perfect interconnection across the network (i.e., copper-plate assumption allows generation to service the load at any node within the same time interval). The net load for each interval of the simulation period is calculated based upon generation power from existing capacity, solar, wind and hydrogen. Solar and wind is used to charge storage when excess energy is available, and storage is discharged for intervals that have an energy deficit. After updating storage values, the spillage for intervals when storage is full and remaining deficit are calculated.

3.2.1 Imports

The numpy package is imported to enable the use of numpy arrays.

3.2.2 Reliability Function

Function that performs the net load calculation over the simulation period. Storage system discharging, charging, and stored energy profiles are generated. Deficit energy and deficit power profiles are also generated.

3.2.2.1 Calculate Net Load for Each Interval

The interval generation power from solar, wind, existing capacity, external interconnections, and hydrogen within a particular solution object are used to balance the load in each interval. A copper plate assumption applies to the energy balance, such that generation from any node can balance demand at any other node. The pumped hydro and battery power and energy capacity from the solution object are defined as variables. Then, numpy arrays are created to store the energy discharged, charged, and stored energies at each interval. Numpy arrays are also created to store the deficit energies and deficit powers at each interval.

$$\begin{aligned} Net\ Load(t) = Demand(t) - Solar\ Generation(t) - Wind\ Generation(t) \\ - External\ Interconnection\ Imports(t) - Existing\ Capacity\ Generation(t) \\ - Hydrogen\ Generation(t) \end{aligned}$$

Positive net load indicates that there is an energy deficit in that interval, while negative net load means that there is excess electricity available that can be stored or spilled.

3.2.2.2 Create Storage System Variables

The power capacity, energy capacity, and efficiency of the storage systems (batteries and pumped hydro) are initialised. Arrays of zeroes for every time interval are initialised to store the profiles for discharging power, charging power, and stored energy for each storage system. Arrays of zeroes for every time interval are also initialised to store the deficit energy and deficit power profiles.

3.2.2.3 Initialise Interval

For each interval in the simulation, storage is dispatched to meet deficit or charged to store excess generation. At the start of an interval, the stored energy is initialised to equal the stored energy at the end of the previous interval (for the first interval of the simulation, stored energy is initialised to equal 50% of total energy capacity of the storage).

3.2.2.4 Update Storage System

If the net load in that interval is positive, then storage is discharged to meet that remaining demand. The discharging power capacity is equal to the minimum of either the net load (enough storage power and energy to meet net load), storage power capacity (not enough storage power capacity to entirely meet the net load), or stored energy divided by the time interval resolution (not enough energy in storage to entirely meet the net load).

$$Discharge(t) = \min\left(max\big(0, Net\ Load(t)\big), Storage\ Power\ Capacity, \frac{Stored\ Energy(t-1)}{resolution}\right)$$

If the net load in that interval is negative, then storage is charged by the excess generation. The charging capacity is equal to the minimum of either the net load (enough storage power and remaining energy storage to charge the excess generation), storage power capacity (not enough storage power capacity to charge all excess generation), or remaining energy capacity divided by efficiency divided by the time interval resolution (not enough space in storage to store all excess generation).

$$Charge(t) = \min\left(-1 \times \min\big(0, Net\ Load(t)\big), Storage\ Power\ Capacity, \frac{Storage\ Energy\ Capacity(t) - Stored\ Energy(t-1)}{efficiency \times resolution}\right)$$

The stored energy for the end of the time interval is updated based upon the charging or discharging behaviour of the storage system.

$$Stored\ Energy(t) = Stored\ Energy(t-1) - Discharge(t) + Charge(t) \times efficiency$$

The net load is updated to `diff1` based upon the charging/discharging behaviour of the first storage technology. The second storage technology charging/discharging behaviour is then evaluated according to the new net load.

3.2.2.5 Determine Deficits

The deficit energy and deficit power are calculated for that time interval, where s is the storage technology. Energy deficits are caused when there is not enough energy in storage to balance demand, and power deficits are caused when the storage system does not have a power capacity high enough to balance demand.

$$Deficit(t) = \max\left(0, Net\ Load(t) + \sum\nolimits_s \big(Charge_s(t) - Discharge_s(t)\big)\right)$$

3.2.2.6 Solution Updates

Simulation.py

The total deficit and spillage for each interval over the simulation period are calculated. Energy is spilled if storage systems are full and cannot charge excess generation from variable sources.

$$Spillage(t) = -1 \times \min\left(0, Net\ Load(t) + \sum\nolimits_s \big(Charge_s(t) - Discharge_s(t)\big)\right)$$

Where s is the storage technology.

3.2.2.7 Error Checking

Constraints on the maximum stored energy, minimum stored energy, minimum deficit, and minimum spillage at each interval are asserted to check for errors.

3.2.2.8 Update Solution Object

The solution object is updated to include the interval discharge capacities, charge capacities, and stored energies. It is also updated to include the interval deficit and spillage. The function then returns the interval deficits and discharged power profiles from the storage systems.

3.3 Network

The "Network.py" module defines the rated capacity of transmission components in the electricity network that was developed in section 2.1 within the FIRM model. It determines the energy imports at each node required to balance demand. Electricity transmission between nodes is then determined based on required imports from one node directly connected to another, as well as transmission entering the directly connected node from other regions.

For each interconnection, the transmission capacity is determined by the maximum power transmitted in any interval. The transmission capacity is used to calculate system costs and energy losses in the "Optimisation.py" and "Statistics.py" modules.

3.3.1 Imports

The numpy package is imported to enable the use of numpy arrays.

3.3.2 Transmission Function

Function that determines the transmission capacity required for each interconnection.

3.3.2.1 Define Interval Node Parameters

The node list, solar site list, wind site list, and external interconnection list are imported from the solution object. The number of intervals and nodes are also imported from the solution object.

Arrays for interval solar generation power, wind generation power, and external interconnection imports for each node in each interval of the simulation are created to store values from the solution object. Once these generation power values are stored in the arrays, the arrays are transposed.

Existing capacity (e.g., storage hydro and biomass) is imported from the solution object. The generation power from existing capacity at each node for each interval is stored in an array and transposed.

The hydrogen capacity at each node is added to a variable, then used to calculate the generation power from hydrogen at each node for each interval within an array that is transposed.

The electricity demand is imported from the solution object and stored in an array.

The deficit, determined by the "Simulation.py" module, is imported from the solution object and scaled according to a `defactor`. The `defactor` variable contains the fraction of total simulation nodal demand within each interval. That is, it is assumed that the simulation deficit is spread across all of the intervals in the simulation in proportion to the size of demand in each interval.

The spillage, determined by the "Simulation.py" module, is imported from the solution object and scaled according to a `spfactor`. The `spfactor` variable contains the fraction of total simulation nodal variable renewable generation (solar and wind) within each interval. That is, it is assumed that the simulation spillage is spread across all of the intervals in the simulation in proportion to the size of variable renewable generation in each interval. If wind and solar generation is 0 at all nodes for a given interval, then spillage is spread evenly across all nodes.

The storage discharged and charged energy, determined by the "Simulation.py" module, is imported from the solution object and scaled according to a `pcfactor` or `bfactor`. The `pcfactor` or `bfactor` variable contains the fraction of the sum of nodal storage power capacity within each interval. That is, it is assumed that charging and discharging energy is spread across all of the intervals in the simulation in proportion to the dispatch power of the storage system in each interval.

3.3.2.2 Required Imports Calculation

The imports required to balance electricity demand for each node (j) in each interval (t) is calculated according to the following formula, where k is the storage system type.

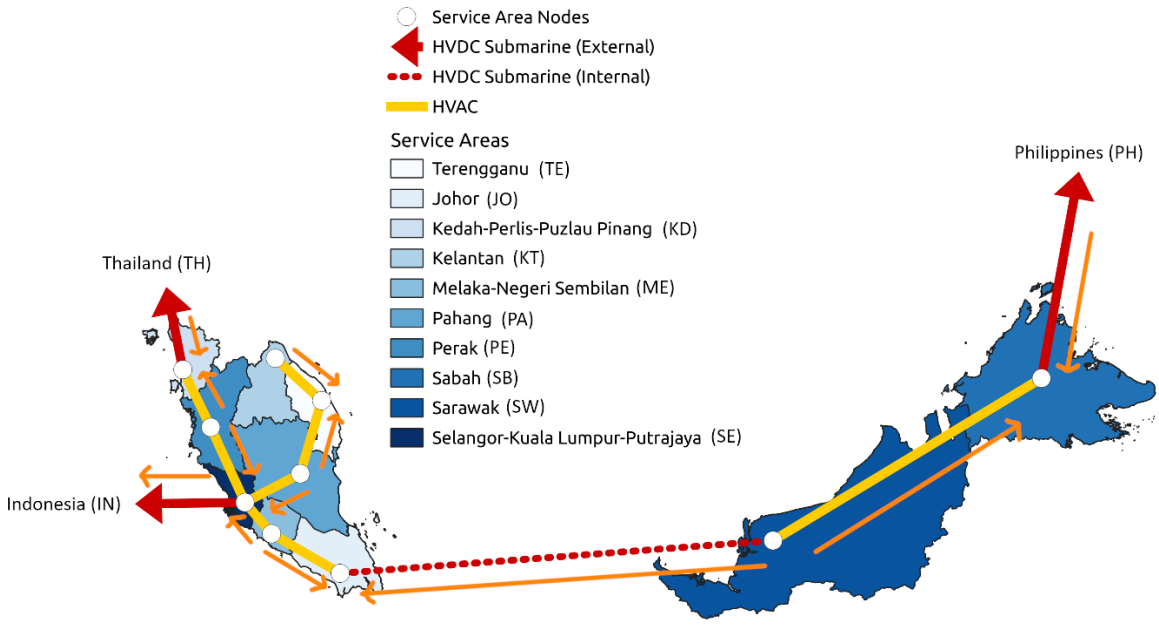
$$\begin{aligned} Required\ Import(t,j) \\ &= Electricity\ Demand(t,j) + \sum_s Charge_s(t,j) + Spillage(t,j) \\ &- Solar\ Generation(t,j) - Wind\ Generation(t,j) - External\ IC\ Imports(t,j) \\ &- Existing\ Capacity\ Generation(t,j) - Gas\ Generation(t,j) - \sum_s Discharge_s(t,j) \\ &- Deficit(t,j) \end{aligned}$$

3.3.2.3 Transmission Capacity

Interconnection unique identifiers are used to name variables that define the transmission capacity for each interconnection. Interconnections each have a sense, which defines the order of nodes forming an interconnection (e.g., Node A to Node B has the opposite sense of Node B to Node A).

An easy way to understand this process is to start by drawing out the transmission network and assigning a sense to each interconnection. This has been done for the Malaysia network in Figure 20. The sense is arbitrary, but must remain consistent throughout the equations.

Figure 20. Malaysia Transmission Network



Each equation defines the power transmitted along an interconnection based upon the necessary imports into a node. The transmission along interconnections for nodes around the outside of the electricity network are defined first. For the Malaysia network, these nodes include PH, TH, IN and KT. Based upon the sense of each interconnection in Figure 20:

$$THKD = -1 \times Required\ Import(t, \mathrm{TH})$$

$$PHSB = -1 \times Required\ Import(t, \mathrm{PH})$$

$$INSE = Required\ Import(t, \mathrm{IN})$$

$$KTTE = -1 \times Required\ Import(t, \mathrm{KT})$$

Transmission along interconnections connected to the next nodes along the network are then defined. Start by looking at the total imports into that node, then subtract off imports of all previously defined transmission into the node.

$$KDPE = Required\ Import(t, \mathrm{KD}) - THKD$$

$$SBSW = Required\ Import(t, \mathrm{SB}) - PHSB$$

$$TEPA = Required\ Import(t, \mathrm{TE}) - KTTE$$

Remember that if the interconnection we are defining has a sense that moves away from our reference import node, then we must multiply the imports by -1.

$$JOSW = -1 \times Required\ Import(t, \mathrm{SW}) - SBSW$$

$$PASE = -1 \times Required\ Import(t, \mathrm{PA}) - TEPA$$

$$PESE = -1 \times Required\ Import(t, \mathrm{PE}) - KDPE$$

We then move one more interconnection along the network again:

$$MEJO = Required\ Import(t, \mathrm{JO}) - JOSW$$

And then to the final interconnection:

$$SEME = -1 \times Required\ Import(t, \mathrm{ME}) - MEJO$$

Note that all of the interconnections have now been defined, but we have not defined any interconnections with reference to node SE yet. This is because there are fewer interconnections than nodes. We can define an interconnection for a second time with reference to node SE to check for any errors in our equations, since both definitions should be equal (or close enough given float accuracy).

$$SEME1 = Required\ Import(t, \mathrm{SE}) - PASE - PESE + INSE$$

$$|SEME - SEME1| < 0.01$$

Note that in the definition for the SEME1 transmission power profile, we subtracted PASE and PESE from the imports into SE, but added INSE. This is because SEME, PASE, and PESE all have a sense that directs them into node SE, while INSE has a sense that directs it away from node SE. So, the net imports into node SE must equal all of the power imported by SEME, PASE, and PESE minus all of the power exported from the node along INSE in each time interval.

`If` statements are used to include or exclude nodes based upon whether they are contained within the node coverage. This can be used for scenarios that contain different electricity network designs. If a node is not contained in the coverage list, a transmission capacity of 0 is assigned to that interconnection for each time interval.

The transmission capacity required for each interval for each interconnection is stored in the `TDC` variable which is returned at the end of the function.

3.3.2.4 Output

If the `output` argument for the transmission function has a value of `True`, then the solution object is updated to include the interval dispatch powers for each technology; interval discharge power, charge power, and energy capacity of storage; and interval deficit and spillage at each node. This is required by the "Statistics.py" module.

3.4 Optimisation

The "Optimisation.py" module runs the least-cost optimisation for the electricity network based upon a Differential Evolution minimisation.[15] By default, the module will also run all other modules required to output the results from the entire FIRM model simulation. Command line arguments for the simulation may be used when running this module to define the scenario or modify the optimisation parameters (e.g., number of iterations).

The Differential Evolution attempts to minimise the objective function based upon the levelised cost of electricity for the network over the simulation period and the penalties for not meeting the reliability, energy, or transmission constraints.

3.4.1 Imports

The `differential_evolution` function is imported from the `scipy.optimise` package. This function is used to find the decision variables that minimise the objective function (i.e., the least cost solution with negligible constraint penalties).

The `ArgumentParser` function is imported from the `argparse` package. This function allows command line arguments to be defined within the module.

The `datetime` and `csv` packages are imported to check simulation times and write csv files containing the optimisation results.

The entire "Input.py" module is imported. The `Reliability` function is imported from "Simulation.py" and the `Transmission` function is imported from "Network.py"

3.4.2 Command Line Arguments

Table 2 contains the command line arguments that are included in the FIRM model by default.

Table 2. Default Command Line Arguments

Parameter	Description	Default Argument	Example Arguments
-i	The maximum number of iterations in the Differential Evolution spent finding a least-cost solution. Larger numbers of iterations are required if there are not enough attempts to reach the global minima.	500	400, 4000
-p	The multiplier for the population size in the Differential Evolution. The population is the set of all solutions	2	2, 10

[15] https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.differential_evolution.html

Parameter	Description	Default Argument	Example Arguments
	tested for a single iteration. Larger population sizes may be required if the Differential Evolution is getting stuck in local minima.		
-m	The mutation constant of the Differential Evolution, also called the differential weight. Increasing the mutation constant increases the search radius, but slows down convergence.	0.5	Float in range [0,2]
-r	The recombination constant of the Differential Evolution, also called the crossover probability. Increasing this value allows a larger number of mutants to progress to the next generation (increasing exploration of solution space), but at the risk of population stability (may compromise local search capabilities).	0.3	Float in range [0,1]
-e	Electricity demand per capita per year scenario. If this argument is included in the model, then the electricity demand inputs should be named "Data/electricity*X*.py" where *X* is the demand per capita per year.	5	5, 10, 20
-n	The name of the network scenario to be simulated. Using the name of a single node will run the simulation for that single node without any interconnections.	Super1	Super1, *node names*, *network scenario names*
-s	The name that specifies any other scenario changes for the simulation. For example, it could be used to specify whether a HVAC or HVDC network is to be used, whether a different cost factor workbook is to be used etc.	HVDC or factor or 1	HVDC, HVAC or factor, factor1 or 1, 2, 3...

3.4.3 Objective Function `F`

The objective function is the function that is minimised in order to find decision variables that produce the least-cost electricity network over the simulation period. A list of potential decision variables is added as an argument (`x`) to the objective function and used to create a solution object based on those decision variables.

Demand is initially balanced by variable renewable energy resources and baseload capacity only. The flexible capacity of each existing capacity and peaking hydrogen technology is then calculated by running simulations that assume these technologies are able to entirely meet the deficit. A simulation with all existing capacity and flexible hydrogen operating at maximum power is used to determine whether deficit across the simulation is within the defined allowance.

Penalty functions are used to set reliability, energy, power, and transmission constraints. A multiplier of 10^6 is applied in each penalty function to make penalties extremely large, ensuring that least-cost solutions have negligible penalties applied.

3.4.3.1 Simulation with Only Baseload

The capacity of hydrogen is imported from the solution object.

The `Reliability` function is called with all of the baseload generation dispatched and no flexible generation dispatched. The total energy deficit for each year is calculated. The maximum interval deficit power from the simulation is used to define the flexible generation power and peaking gas required (`PFlexible_Gas`).

3.4.3.2 Simulation with Baseload and Existing Capacity

The `Reliability` function is called repeatedly with all of the baseload generation dispatched, adding in each flexible generation technology from cheapest to most expensive. The total energy deficit for each year is calculated. The maximum interval deficit power from the simulation is used to define the remaining flexible generation power required.

3.4.3.3 Generation from Existing Capacity

Within each technology, the maximum annual generation would occur when storage systems are pre-charged by the existing capacity (due to efficiency losses of the storage systems). Therefore, the maximum annual generation assumes each storage technology is pre-charged in proportion to the total power discharged by the technology within the simulation.

The generation energy for existing capacity is calculated based upon the maximum difference between interval deficits from subsequent `Reliability` simulations. For example, assume there is a system with existing hydroelectricity and hydrogen. The Simulation with Only Baseload runs a simulation with only baseload hydroelectricity dispatched from the existing capacity. The Simulation with Baseload and All Existing Capacity runs a simulation with baseload hydroelectricity, all flexible hydroelectricity, and no hydrogen. The hydroelectric generation energy is calculated based upon the maximum difference between the annual deficit energy from the second simulation and corresponding annual deficit from the first simulation (divided by the efficiency of the storage systems to assume all flexible power pre-charges storage in the worst-case scenario). The hydrogen generation capacity required to meet the remaining deficit is equal to the maximum annual deficit from the second simulation (also divided by the efficiency of the storage).

3.4.3.4 Power and Energy Deficit Functions

An energy penalty function is used to constrain the generation energy of all existing capacity and peaking gas within the defined maximums.

$$Energy\ Penalty = \max(0,10^6 \times \left(\sum_{t=0}^{n} \left(Generation\ Energy\ (i,t) - Maximum\ Allowable\ Generation(i)\right)\right)$$

where *i* is the existing capacity or peaking gas technology.

A power penalty function is used to constrain the generation power of all existing capacity and peaking gas within the predefined maximums.

$$Power\ Penalty = \max(0,10^6 \times \left(\sum_{t=0}^{n} \left(Generation\ Power\ (i,t) - Maximum\ Allowable\ Power(i)\right)\right)$$

where *i* is the existing capacity or peaking gas technology.

3.4.3.5 Deficit Penalty Function

The `Reliability` function is called with all of the baseload generation dispatched, all existing flexible generation dispatched, and all hydrogen dispatched.

A deficit penalty function is used to constrain the deficit within the pre-defined deficit allowance.

$$Deficit\ Penalty = \max(0,10^6 \times \left(\sum_{t=0}^{n} \left(Deficit(t)\right) - allowance)\right)$$

3.4.3.6 Existing Capacity Generation Profiles

A rough generation profile for each existing capacity technology and peaking gas technology is created by clipping the deficit profiles from the corresponding simulations to be within 0 and the total capacity of that technology.

3.4.3.7 Transmission Penalty Function

A final `Reliability` simulation is run using the final generation profiles for existing capacity and peaking gas technologies. The average annual discharged energy from the storage systems during that simulation is calculated.

The transmission capacity for each interconnection in the network required to import power necessary to balance demand with generation is calculated using the `Transmission` function. A transmission penalty function is used to constrain the capacity of certain interconnections within pre-defined maximums.

$$Transmission\ Penalty = \max(0,10^6 \times \left(\sum CT_k - CTMax_k\right)$$

For transmission line *k*.

3.4.3.8 Levelised Cost of Electricity

The average annual generation from each existing capacity and peaking gas technology is calculated based upon the deficits from the corresponding simulations, accounting for the efficiency of the storage to assume all flexible generation pre-charges storage in the worst-case scenario. Average annual electricity imported through external interconnections is also calculated.

The annualised cost of all generation, transmission, and storage technologies is calculated based upon the cost factor inputs. Transmission losses for each interconnection are calculated and used to determine the average annual transmission losses. The levelised cost of electricity is then calculated based upon the annual cost and energy in the network.

$$LCOE = \frac{Cost_{solar} + Cost_{external\ IC} + Cost_{wind} + Cost_{transmission} + Cost_{storage} + Cost_{existing\ capacity}}{Annual\ Electricity\ Demand - Annual\ Transmission\ Losses}$$

3.4.3.9 Function Result

The total penalty and levelised cost of electricity for each optimisation iteration is saved as a new row to "Results/record.csv". The objective function is then calculated based on the levelised cost of electricity and penalties.

$$F(x) = LCOE + Deficit\ Penalty + Energy\ Penalty + Transmission\ Penalty + Power\ Penalty$$

3.4.4 Main Block

If the "Optimisation.py" module is run from the command line, then it will run the main block. This block performs the Differential Evolution optimisation and then runs the dispatch or fill analysis.

3.4.4.1 Differential Evolution

The lower bound and upper bound for each decision variable in the Differential Evolution are defined in the `lb` and `ub` variables respectively. A Differential Evolution of the objective function with the defined upper and lower bounds for the decision variables is then run. Command line arguments can be used to modify the Differential Evolution parameters, which may be necessary for finding the global minimum of complex networks.

3.4.4.2 Results

The optimised decision variables are written to "Results/optimisation_resultx_*scenario*.csv". The optimisation time is printed to the terminal.

3.4.4.3 Dispatch Analysis

By default, the FIRM model will run the `Analysis` function from the "Fill.py" module after completing the optimisation. Older versions import the `Analysis` function from the "Dispatch.py" module instead.

3.5 Dispatch (Deprecated)

The "Dispatch.py" module has been replaced by the "Fill.py" module in more recent versions of the FIRM model.

The "Dispatch.py" module determines the actual generation profiles for flexible existing capacity and peaking gas. The profiles for each year in the simulation period are calculated in parallel. For long simulation periods, the "Dispatch.py" module may take an extremely long time to complete, so it may be necessary to use the "Fill.py" module to generate these generation profiles instead.

During the optimisation, the deficit was expected to be entirely met by flexible generation. The interval deficit profile may exceed the available hydro capacity, though, since it may be necessary for the flexible generation

to pre-charge storage systems to make it through these high deficit intervals. Therefore, the generation profiles must be formed on an hour-by-hour basis.

3.5.1 Imports

The `Pool` function and `cpu_count` variable are imported from the `multiprocessing` package to allow parallel processing of each dispatch year within the simulation period.

The `datetime` package is imported to calculate the time required to run the dispatch.

Everything from the `Input` module is imported and the `Reliability` function is imported from the Simulation module.

3.5.2 Flexible Function

3.5.2.1 Initialisation

The start and end intervals for the dispatch year are defined (since generation profiles for each year in the simulation period are generated in parallel).

The flexible capacity and peaking gas profiles are initialised to operate at maximum capacity for the entire simulation period, resulting in a deficit smaller than the allowance (refer simulation in section 3.2).

3.5.2.2 Profile Builder

An iteration on each time step within the dispatch year is performed. At each interval, the existing capacity and peaking gas is re-initialised to its baseload capacity for that interval (e.g., run-of-river hydro, no hydrogen etc.). A simulation is run to determine whether there is a deficit in that year after zeroing the flexible generation.

If the deficit is smaller than the allowance, then the flexible generation is turned off for that interval. Otherwise, flexible generation is turned on starting with the cheapest flexible generation technology first. Once the deficit is smaller than the allowance, the loop moves to the next iteration.

The generation profile returned by the loop defines interval generation to be either 0 MW or maximum capacity. These profiles are then clipped based on the spillage in each interval, giving a value **between** 0 MW and maximum capacity for each interval.

3.5.3 Analysis Function

3.5.3.1 Multiprocessing

The pool of workers is created, and a set of instances is created, to define each year of dispatch within the simulation period. The pool of workers is assigned to run the `Flexible` function for each dispatch year. After all tasks are completed, the pool is terminated.

3.5.3.2 Results

The generation profiles for each flexible generation technology are saved to the "Results/Dispatch_*Technology*.csv" files.

The time to complete the dispatch tasks is printed into the terminal.

By default, the `Information` function from the "Statistics.py" module is run at the end of the dispatch.

3.6 Fill

The "Fill.py" module is a replacement for the "Dispatch.py" module in order to reduce processing time for either large simulation periods or FIRM models with many flexible generation technologies. The "Fill.py" module uses a deficit targeted model to determine generation profiles for flexible existing capacity and peaking gas technologies.

The power of all flexible technologies is initialised to equal their maximum capacity, other than the most expensive technology which is initialised to output zero power over the simulation. A `Reliability` simulation is run to determine the deficit that must be covered by the most expensive flexible generation technology.

The deficit at each time interval is filled with the most expensive flexible generation technology. If the deficit exceeds the capacity of that flexible generation technology, then the power of the flexible generation technology is increased in previous time intervals to pre-charge storage (subject to capacity and annual generation constraints for that technology); the storage is then used to meet the deficit gap.

The next most expensive flexible generation technology is then set to have zero power over the simulation. The generation profile for the most expensive generation technology and the maximum power profiles for all other flexible generation technologies are used to run the `Reliability` simulation. Deficit at each time interval is filled with the second most expensive flexible generation technology, with storage being pre-charged by that technology if the deficit exceeds maximum capacity for that technology in any given time interval.

This process is repeated until all generation profiles for flexible existing capacity and peaking gas technology have been created.

3.6.1 Imports

The `numpy` package is imported to save data to CSV files and handle array functions.

Everything from the `Input` module is imported and the `Reliability` function is imported from the Simulation module.

3.6.2 Fill Deficit Function

The `fill_deficit` function is used to fill the deficit for a specific simulation with generation from a particular flexible generation technology. Flags are used to identify the flexible generation technology that is being used to fill deficit when the function is called.

3.6.2.1 Initialise

Determine all time intervals where there is an energy deficit greater than 0 and store them in the `idx` variable.

Iterate through each time interval, assigning the deficit at that interval to `d` and initialising the interval number `t` to be the first interval with a remaining deficit greater than 0 (`i`). For each interval try to fill the deficit with the specific flexible generation technology.

3.6.2.2 Deficit Filling While Loop

Form a `while` loop that iteratively fills the deficit in interval `i`, breaking the loop when there is either no more interval deficit, annual generation for the flagged technology has been reached, or the number of iterations reaches the `step` limit. Start at the first interval where there is remaining deficit in the simulation (`t=i`). Determine the first interval and last interval in the simulation year containing `t`. The year is used to confirm annual generation limits for each technology are not exceeded.

If `t = i-1`, then recalculate the interval deficit to account for the storage system efficiency (since storage is required to meet deficit when there is not enough flexible generation capacity available in that time interval). The lowest efficiency storage system is currently assumed to be pre-charged to account for the worst-case scenario.

Fill the deficit based upon the flexible generation technology flags that have a `True` value. Only one flag should be active for a given function call. Calculate the remaining annual generation available for that flexible generation technology. If there is no remaining annual generation for that technology, raise an exception that breaks the `try` statement.

Calculate the generation capacity for the flagged flexible technology dispatched for interval `t` to meet deficit. The generation capacity is the minimum of either the current profile value plus deficit (enough capacity to entirely meet deficit in that interval), the maximum capacity of that technology (not enough power to meet deficit in that interval), or the current profile value plus the remaining annual generation (not enough annual energy to meet deficit in that interval). The remaining interval deficit is then calculated once flexible generation from the flagged technology has been dispatched. The flagged technology generation profile is updated at interval `t`.

If the annual generation limit for the flagged technology has been reached, break the `while` loop. Otherwise, redefine `t` to be the most recent interval where the flagged technology is not currently dispatching at maximum capacity. The while loop will then iterate again to increase the capacity in this previous interval in an attempt to pre-charge storage to meet the deficit gap in interval `i`. The number of intervals where pre-charging can occur is limited by the `step` argument.

Once the deficit for the entire year has been filled, return the generation profiles for each flexible generation technology.

3.6.3 Save Function

Save the generation profiles for each flexible generation technology to the "Results/Dispatch_*Technology_scenario*.csv" files.

3.6.4 Mean and Max Functions

The `mean` function calculates that average annual generation for a given flexible generation technology. The `maxx` function determines the maximum interval generation for a given flexible generation technology.

3.6.5 Main Block

Performs the deficit fill operation for each flexible generation technology when the "fill.py" module is run from the command line.

3.6.5.1 Initialise

The decision variables from the least-cost solution, generated by the "Optimisation.py" module, are imported and used to create a solution object.

3.6.5.2 Preliminary Simulations

Preliminary simulations are performed in order to determine which flexible generation technologies are required to meet deficit over the entire simulation period.

The first simulation involves only baseload generation being dispatched to meet deficit (i.e., no flexible generation). The average annual deficit for the first simulation is calculated. The flexible generation power required to meet the maximum interval deficit power, `PFlexible_Gas`, is calculated.

The second simulation dispatches the cheapest flexible generation technology at maximum power to cover deficit. Remaining average annual deficit for the second simulation is calculated. The flexible generation power required to meet the maximum remaining interval deficit power is calculated.

This process can be repeated for all flexible generation technologies.

The maximum annual generation energy for all existing capacity and peaking gas technologies is calculated based upon the difference in annual deficits for consecutive simulations (accounting for pre-charging of the lowest efficiency storage system to account for the worst-case scenario).

3.6.5.3 Deficit Filling

Deficit filling starts with the most expensive flexible generation technology. If the maximum annual generation energy for that technology in section 3.6.5.2 is zero, then skip that technology.

Initialise all flexible generation technologies to operate at maximum capacity for the entire simulation. Initialise the most expensive flexible generation technology to output 0 power for the entire simulation. Calculate deficit that must be filled by the most expensive flexible generation technology using the `Reliability` function. Then, create a generation profile for the most expensive flexible generation technology using the `fill_deficit` function. Continue calling the `fill_deficit` function using the most recent generation profiles until the deficit over the entire simulation period has been met with the most expensive flexible generation technology (up to a maximum of `step` repetitions).

Proceed to the second most expensive flexible generation technology. Redefine the second most expensive generation technology to output 0 power for the entire simulation (i.e., baseload dispatched, cheaper flexible generation dispatched at maximum power, and more expensive flexible generation dispatched based upon previously determined generation profiles). Calculate deficit that must be filled by the second most expensive flexible generation technology using the `Reliability` function. Then, create a generation profile for the second most expensive flexible generation technology using the `fill_deficit` function. Continue calling the `fill_deficit` function using the most recent generation profiles until the deficit over the entire simulation period has been met with the second most expensive flexible generation technology (up to a maximum of `step` repetitions).

Repeat this process until generation profiles have been created for all flexible existing capacity and peaking gas technologies.

Save the generation profiles using the `save` function.

3.7 Statistics

The "Statistics.py" module checks the feasibility of the solution and exports energy balance profiles and statistics. It generates the load profile and generation mix data (LPGM) and the storage and transmission information (GGTA) based upon the decision variables for the least-cost solution found by the "Optimisation.py" module. The flexible generation profiles from the "Dispatch.py" or "Fill.py" modules are also required.

3.7.1 Imports

The `datetime` package is imported to check time taken to run the module.

The `numpy` package is imported to enable the use of arrays.

The entire "Input.py" module is imported. The `Reliability` function is imported from "Simulation.py" and the `Transmission` function is imported from "Network.py"

3.7.2 Debug function

The `Debug` function is used to check that the results stored in the solution object do not exceed the limits of the system. If exceptions are thrown by the Debug function, it indicates that there is a mistake in the calculations performed in a previous module.

3.7.2.1 Initialise Result Values

Import the load, generation, storage, deficit, and spillage results from the solution object and store them in respective variables.

3.7.2.2 Interval Debugging

For each interval within the simulation period, assert the following information:

- The NetLoad is approximately 0 (noting that floating point errors may mean it is not exactly 0);
- The energy stored in each storage system equals the stored energy from the previous interval minus the discharged energy plus the charged energy for that interval;
- The power dispatched by each generation/storage technology is less than the maximum capacity of that technology.

3.7.3 LPGM Function

The load profiles and generation mix (LPGM) function generates a file that shows the energy balancing for the network and each node for every interval of the simulation.

3.7.3.1 Generate Network-level LPGM

Run the `Debug` function to check for errors in the solution.

Join the load, generation, storage, and transmission profile arrays. Transpose the matrix to produce an array where each column represents a particular profile and each row represents an interval in the simulation. Round the results to the nearest integer.

The datetime of each interval is determined and inserted as the first column into the results matrix.

The header row for the results file is defined to match the order of columns in the results matrix.

Network-level results are saved to the "Results/LPGM_*scenario*_Network.csv".

3.7.3.2 Generate Node-level Results

If the scenario includes a multi-node network, then the LPGM results are generated at a nodal level too.

The header row for the node-level results is defined to match the order of columns in the nodal results matrix.

The `Topology` profile matrix for the scenario's nodal coverage is defined. The `Topology` variable is defined in the `Information` function. It specifies the sum of interval imports along interconnections into each node of the network.

For each node in the network, create a node-level results matrix that demonstrates the energy balancing for each interval of the simulation. Transpose the matrix and round the results to the nearest integer.

Add a datetime column to the node-level results matrix and save the results for that node to "Results/LPGM_*scenario_node*.csv". Repeat for all nodes in the network.

3.7.4 GGTA Function

The GW, GWh, TWh p.a. and A$/MWh (GGTA) function summarises the results at a network-level for the entire simulation period.

3.7.4.1 Import Cost Factors

Import the cost factor data from the "Data/factor.csv" file and convert into a dictionary object. The `scenario` variable can be used to select different cost factor data.

3.7.4.2 Import Capacities from Least-Cost Solution

The network-level power capacity (GW) of each generation and storage technology is calculated based upon the data in the solution object or variables defined in the "Input.py" module. The energy capacities (GWh) of the storage technologies are also imported from the solution object.

3.7.4.3 Import Generation Energy from Least-Cost Solution

Calculate the average annual generation (GWh) for each generation technology and discharged energy (GWh) for each storage system. Determine the average annual capacity factor for the wind and solar generators.

3.7.4.4 Calculate the Annual Costs for Each Technology

Recall that the cost factors ($/W/year) are based upon the total lifetime costs and a discount rate, originally defined within the " Results/Calculations.xlsx" workbook.

Calculate the cost ($b per year) of each generation and storage technology. New generation capacity costs (solar, wind, and hydrogen) are based upon the capital costs, operation and maintenance costs, and fuel costs. The cost of existing capacity is based upon the levelised cost of electricity for each technology in that region. The cost of imports from external interconnections should be based upon known or estimated prices of electricity imported from external regions.

The cost of transmission is split into DC and AC costs. The DC costs include the cost of all HVDC interconnections (capital costs and O&M costs of lines and converter stations, as included in the cost factor). The AC costs include the cost of all HVAC interconnections (capital costs and O&M costs of lines, as included in the cost factor) plus the AC costs required to connect wind and solar power stations to the grid. Note that the cost factors in the "Results/Calculations.xlsx" workbook assume that solar is 10 km from transmission and wind is 40 km from transmission by default.

3.7.4.5 Calculate the Average Annual Energy Demand

Determine the average annual electricity demand (PWh/year) and the average annual transmission losses (PWh/year) for the simulation.

3.7.4.6 Calculate the Levelised Cost of Electricity

Using the annual cost of each technology, average annual electricity demand, and average annual transmission losses, the levelised cost of electricity (LCOE) is calculated at a network-level. The LCOE contribution from each technology is also calculated.

The levelised cost of electricity can be separated out into two components:

$$LCOE = Levelised\ Cost\ of\ Generation + Levelised\ Cost\ of\ Balancing$$

The levelised cost of generation (LCOG) is the cost of generating all electricity within the network, regardless of spillage and energy losses. The levelised cost of balancing (LCOB) is the cost associated with energy storage, transmission infrastructure, spillage, and transmission or efficiency losses.

3.7.4.7 Calculate the Levelised Cost of Generation

The total LCOG is calculated using the costs at average annual generation for all generation technologies. LCOG for each individual technology is also calculated. Note that the total LCOG does not equal the sum of LCOGs for each individual technology. Rather, it is a weighted average of the LCOGs.

3.7.4.8 Calculate the Levelised Cost of Balancing

The total LCOB is calculated based upon the difference between LCOE and LCOG. The individual contributions to LCOB of storage system costs, transmission costs, and spillage & energy losses is then calculated.

3.7.4.9 Print Levelised Costs

Key levelised costs are printed into the terminal.

3.7.4.10 Save Results

An array of results is created to store the average annual demand (TWh), average annual transmission losses (TWh), generation technology power capacities (GW), generation technology average annual generation (GWh), storage technology power capacity (GW), storage technology energy capacity (GWh), interconnection transmission capacities (GW), LCOE, LCOG, and LCOB.

The array of results is then saved to the "Results/GGTA_*scenario*.csv" file.

3.7.5 Information Function

The `Information` function prepares the results from the optimal solution and calls the `LPGM` and `GGTA` functions to save the results.

3.7.5.1 Initialise the Solution and Check Simulation Results

A solution object is initialised using the decision variables from the least-cost optimisation performed in the "Optimisation.py" module. Run the simulation using the `Reliability` function, inputting the optimal decision variables (from "Optimisation.py") and the flexible generation profiles (from "Fill.py" or "Dispatch.py").

Assert that the deficit from the simulation is roughly zero (noting that floating point errors may cause small amounts of deficit). Assert that the maximum annual generation from each existing generation or peaking gas technology is lower than the specified annual energy constraints.

Determine the interconnection transmission flows and required capacities using the `Transmission` function. Assign the transmission power flow profiles to each interconnection in the solution object.

If the simulation is for a single node, then define the matrices in the solution object used for the energy balancing. These matrices are otherwise defined within the `Transmission(S, output=True)` function call.

The flexible generation profiles are clipped to their maximum capacity of each flexible generation technology.

The energy capacity for each storage technology is apportioned between nodes according to the fraction of storage system power capacity at each node.

The `Topology` variable is defined. It specifies the sum of interval imports along interconnections into each node of the network. The sense of each interconnection is the same as that defined in section 3.3.2.3.

3.7.5.2 Produce Results

Call the `LPGM` and `GGTA` functions with the solution object. Print the time taken to run the entire "Statistics.py" module.

4 Further Reading

[1] C. Cheng, A. Blakers, M. Stocks and B. Lu, "100% Renewable Energy in Japan," *Energy Conversion and Management,* vol. 255, no. 115299, 2022.

[2] A. Blakers, B. Lu and M. Stocks, "100% renewable electricity in Australia," *Energy,* vol. 133, pp. 471-482, 2017.

[3] B. Elliston, I. MacGill and M. Diesendorf, "Least cost 100% renewable electricity scenarios in the Australian National Electricity Market," *Energy Policy,* vol. 59, pp. 270-282, 2013.

[4] B. Lu, A. Blakers, M. Stocks and T. N. Do, "Low-cost, low-emission 100% renewable electricity in Southeast Asia supported by pumped hydro storage," *Energy,* vol. 236, 2021.

[5] C. Cheng, N. P. Gutierrez, A. Blakers and M. Stocks, "GIS-based solar and wind resource assessment and least-cost 100 % renewable electricity modelling for Bolivia," *Energy for Sustainable Development,* vol. 69, pp. 134-149, 2022.